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**Automatic Extraction of Speech Biomarkers for the Early
Diagnosis of Neurocognitive Disorder**

Final Dissertation in:

Natural Language Processing

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Abstract

The global prevalence of dementia (clinically categorized as Major Neurocognitive Disorder) is expected to rise as the ageing population increases, foreshadowing an immense impact on healthcare systems and societies worldwide. While there is no effective treatment for the illness, early intervention is crucial for slowing down its progression. Recent evidence shows that linguistic alterations are among the first effects of neurocognitive disorder, which has led to the ambition of creating automatic, non-invasive screening systems utilizing digital linguistic biomarkers.

This study aims to explore the predictive potential of acoustic biomarkers for the accurate detection of Mild Cognitive Impairment (MCI) from Italian speech, with the creation of an enhanced acoustic feature set with respect to previous works. To this aim, an automatic feature extraction pipeline was designed, for the extraction of prosodic, spectral and voice quality correlates.

Utilizing the pipeline, the acoustic features were extracted from the speech samples of a corpus comprised of individuals diagnosed with Mild Cognitive Impairment (MCI) and a healthy control group. For a comprehensive analysis, the relevance of each feature was evaluated for its predictive and monitoring potential, also taking into account the linguistic tasks used to elicit the speech samples.

The statistical analysis performed to assess the relevance of each feature for discriminating between the two groups demonstrated high significance for the majority of the features extracted. This study contributes to the exploration of digital linguistic biomarkers for the automatic identification of early signs of the disease, whilst also providing insights towards future directions.

Keywords: Neurocognitive disorder, Mild Cognitive Impairment, Natural Language Processing, Digital Linguistic Biomarkers, Acoustic Feature Extraction

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1. Introduction

1.1 Context and Motivation

In recent years, the global prevalence of dementia has risen dramatically, with current estimates approximating over 55 million cases worldwide, and nearly 10 million new cases arising each year. As the ageing population increases, these numbers are expected to grow even more rapidly, projecting a massive impact on healthcare systems and societies worldwide. The neurodegenerative processes leading to dementia (clinically categorized as Major Neurocognitive Disorder) result to a long-term cognitive deterioration whose severity increases over time, with the average life expectancy estimated at 7 years after diagnosis.

At present there is no effective treatment for the illness, but interventions aiming towards controlling its progression are shown to be more effective in the initial stages of the disease. Thus, medical professionals and researchers stress the importance of early diagnosis, but this attempt can be hindered by strained healthcare systems, high cost of screening examinations, as well as limitations in the traditional diagnostic measurements.

Given considerable evidence that linguistic alterations are among the first effects of neurocognitive disorder, manifesting up to 20 years before patients exhibit observable symptoms, Natural Language Processing research has expanded into the medical field, with the objective to devise reliable, non invasive screening systems based on linguistic analysis. Utilizing linguistic biomarkers, namely acoustic, lexical or semantic features of speech that may capture cognitive impairments, automatic systems trained to identify affected individuals from healthy controls are showing ever-increasing performance.

Although this is very promising, scholars stress the importance of prioritizing risk prediction, advocating for more studies specifically applied to the early stages of the illness, a prodromal phase (Mild Cognitive Impairment) whose linguistic patterns can be more challenging to discern from those of healthy ageing. While there is an ever-growing body of literature devoted to this objective, many of the research efforts focused on non-English languages are still in their initial stages. This presents an important limitation as typological differences among languages cannot allow the generalisability of findings. Additionally, possible variations among the computational approaches implemented between studies may present a barrier for the reproducibility of findings.

1.2 Objectives

The future clinical applicability of this domain necessitates ongoing investigation within the context of each language. To address this need, this thesis inquires about **the discriminative potential of acoustic features** for detecting Mild Cognitive Impairment in Italian. Acoustic features in particular offer the advantage of being independent from textual transcriptions, whose procurement requires additional processing, often necessitating time-intensive manual evaluation.

Considering the current research demands, this work has the following objectives:

1. the creation of an **automatic feature extraction pipeline** implementing an **enhanced acoustic feature set** with respect to previous works, that targets the early signs of cognitive impairment and integrates **standardized** speech processing tools, to **ensure reliable measurements** and **meaningful comparisons** across studies;
2. the **comprehensive evaluation** of the proposed feature set on Italian samples, in order to gain insights about the potential of acoustic variables for detecting and monitoring the disease.

1.3 Thesis structure

This work is comprised of five parts:

1. **Introduction:** this first section outlines the context and objectives of this thesis.
2. **Background:** provides more information about neurocognitive disorders and their detection through acoustic components of speech, discussing the current findings on acoustic features and limitations, which leads to the design of the proposed approach.
3. **Methodology:** describes the approach implemented for addressing the aims of this thesis and the Italian corpus utilized, and a detailed description of the acoustic feature extraction package.
4. **Analysis:** presents the evaluation of the acoustic features for their power in the identification and monitoring of Mild Cognitive Impairment in Italian, and discusses the implications of findings compared to previous works.
5. **Conclusion and Future Work:** summarizes the findings of this thesis and suggests new directions towards the study of acoustic biomarkers for the automatic detection of Mild Cognitive Impairment.

2. Background

2.1 Neurocognitive Disorders

2.1.1 Dementia (Major Neurocognitive Disorder)

Dementia (Major Neurocognitive Disorder) is a category of neurodegenerative diseases that entails a long-term and progressive decline of cognitive functioning. It is most commonly associated with symptoms such as memory loss, deteriorating cognitive-executive functions, impairment of language and motor skills, as well as general emotional distress. These symptoms are the results of various conditions that affect the brain, predominantly Alzheimer's Disease (causing 60 to 70% of dementia cases), cerebrovascular disease, and other brain conditions, including Parkinson's disease, frontotemporal dementia and stroke. Throughout the illness, the severity of cognitive decline and its outcomes increases, reducing the patient's autonomy and wellbeing, as well as their caregivers.

The World Health Organization estimates more than 55 million individuals with Major Neurocognitive Disorder worldwide, with over 60% in low-and middle-income countries, and nearly 10 million new cases arising every year, bringing a vast pressure on healthcare systems. Dementia is currently the seventh leading cause of death and one of the major causes of disability among the elderly. After diagnosis, the average life expectancy is around 7 years, with less than 3% patients living longer than 14 years.¹

2.1.2 Mild Cognitive Impairment (Mild Neurocognitive Disorder)

Mild Cognitive Impairment (MCI) or Mild Neurocognitive Disorder (MND) is used to describe a modest decline of cognitive abilities, considered to be an intermediate state

¹<https://www.who.int/news-room/fact-sheets/detail/dementia>

between normal cognition and Major Neurocognitive Disorder (Morris, R. G. 2008). While dementia is typically diagnosed when the level of cognitive impairment significantly compromises one's abilities to perform essential tasks, individuals living with MCI can remain independent when performing daily living activities, albeit with greater effort. (Hugo, J. & Ganguli, M. 2014).

Systematic reviews and meta-analyses report that over 46% of individuals with MCI progress to clinical dementia within 3 years, and stress that the global prevalence of MCI is augmenting rapidly every year, with the current estimate being up to 19.7% (Song et al., 2023). Compared to cognitively healthy individuals, subjects with MCI exhibit a more severe deterioration rate of cognitive functioning and a mortality rate.

2.1.3 Risk Factors

The highest risk factor associated with cognitive impairment is age, and the constant increase of population over 65 years old worldwide foreshadows an immense societal impact of dementia care in the following years. At present, there is no effective treatment for cognitive impairment, but current interventions aiming to slow down its progression (pharmacological or activities related to cognitive skills for daily living) have been shown to be more effective in the early stages of the disease (Anderson, 2019). A timely diagnosis is crucial for extending cognitive functioning that can delay cognitive deterioration (Poptsi et al., 2022), and can also aid in controlling its psychiatric manifestations (Rebeira et al., 2003).

Mild Cognitive Impairment is considered as an “intervention window” to delay the onset of more serious symptoms of neurodegeneration, which makes early diagnosis essential for developing effective strategies for disease progression. The medical and academic research community have put considerable efforts on the early detection of neurocognitive disorder, but to this day, healthcare systems present many barriers for obtaining an accurate and timely diagnosis, with studies on EU countries reporting that the overall mean time between the first alarming symptoms and receiving a diagnosis can be up to 2.1 years (Dumas et al., 2023). It is therefore of utmost importance to develop non-intrusive and reliable measurements that can aid the diagnostic process.

2.1.4 Neurocognitive disorder and language

While memory loss is usually regarded as the most prominent symptom of dementia, alterations in speech and language are also common, and represent a fundamental marker of cognitive deterioration. Speech is a complex multistage process that converts conceptual ideas into acoustic signals, which involves the parallel processing of different linguistic components, such as the selection of words and proper morphological forms, syntactic planning, which are needed for communicating intent. Affected individuals exhibit slight linguistic deficits at the initial stages of the disease, which become more severe as the rate of cognitive impairment progresses. Hence, traditional diagnostic measurements of language functions are used to detect and monitor the stages of neurocognitive disorder (Szatloczki et al., 2015).

2.1.5 Clinical assessment

The most widely used, non-intrusive screening devices are designed for assessing an individual's cognitive state with prompts that generate language expressions that can hint in their ability to register, recall and effectively communicate information, among other tasks. The most prevalent methods for gathering connected speech involve (semi-)spontaneous tasks, mainly describing images, narrating stories, and engaging in conversation, to replicate natural discourse.

The most frequently applied cognitive test is the Mini-Mental State Examination (MMSE), introduced in 1975 (Folstein et al., 1975), which over time inspired the creation of a variety of different tests with similar tasks. While its validity as a screening tool for cognitive impairment is supported by extensive empirical evidence, it has considerable disadvantages: lack of standardization, lack of suitability for illiterate subjects, inadequate exploration of all cognitive domains, the effect of socio-educational variables on the results (Gallegos et al., 2022). Most importantly, it shows a low sensitivity for differentiating between MCI and early stages of dementia (Delavaran, 2017; Martínez-Nicolás et al., 2022), which are better assessed using a similar screening tool, the MoCA test (Montreal Cognitive Assessment), designed with high sensitivity for detecting MCI (Nasreddine et al., 2005).

2.1.6 Linguistic biomarkers for the detection of neurocognitive disorder

Findings have established that individuals diagnosed with neurocognitive disorder exhibit lexical impairments, naming and word-finding difficulties (anomia), as well as difficulty accessing semantic information intentionally. This leads to decreased verbal fluency, inadequacies at the morphosyntactic level (inflection errors or incomplete sentences), and reduced cohesion at the pragmatic level, i.e. deficits in forming relevance within/between sentences, or overall coherence (Boschi et al., 2017; Mueller et al., 2018).

The effect of neurocognitive disorders across various cognitive domains is known to start silently up to 20 years before an individual shows observable cognitive symptoms (Dubois et al., 2015). This is where the field of Natural Language Processing (NLP) plays an instrumental role. The constant technological advances have made possible to study multiple aspects of speech, many of which are often not perceivable by the human ear. With the automatic examination of different linguistic features, Natural Language Processing can provide an objective way to assess cognitive changes; currently, there is promising research in this field with the scope to devise reliable and non-intrusive tools for the automatic detection of cognitive impairment through speech, which can eventually establish more accessible alternatives for determining whether an individual should seek further professional advice.

Linguistic variables, also known as features, are measures used to describe quantitative and qualitative aspects of language production. Similarly to how the medical field denotes symptoms that represent biomarkers of a disease, the term Digital Linguistic Biomarkers was coined for the study of linguistic variables associated with certain pathologies in the NLP domain, with the aim of providing reliable predictive factors that can contribute to an early diagnosis.

State of the art frameworks consider all the quantifiable aspects of speech in order to identify the most significant features related to neurocognitive disorders, such as acoustic, lexical or semantic variables that capture linguistic deficits. These are leveraged for training automatic systems for distinguishing cognitively impaired individuals from healthy control groups. Since this work focuses on the acoustic biomarkers of early stage neurocognitive disorder, the most relevant acoustic features will be discussed.

2.2 Basic introduction to signal processing

Before describing the acoustic features employed for identifying neurocognitive disorder through speech analysis, it is important to demonstrate how speech is processed and represented for a voice related task - e.g. speech synthesis, automatic transcription, and the extraction of digital linguistic biomarkers. This short description serves to introduce solely basic concepts which are important for understanding both the variety of features used for speech analysis, and the methodology of this project that will be discussed in the following chapters.

2.2.1 Signal processing

When a speech -or any sound- sample is obtained and stored in a digital system, it is represented as a finite set of values (samples) in the time domain, where each sample represents the amplitude (intensity) of the signal at a given time. This discretization is necessary, since digital systems can only process and store a finite number of values with a finite precision, and a series of steps is done in order to ensure that minimal errors are introduced when transforming the continuous signal to its finite representation. After that initial processing, the signal is represented as an array of samples (integers) in the time domain.

Signal attributes change over time, and a meaningful analysis requires the estimation of these attributes periodically, in overlapping segments, instead of performing a single global analysis over the whole signal or its derived spectrum. Hence the first step for analyzing speech samples involves splitting the signal into short, overlapping segments (framing), since speech signals are inherently not stationary, but their processing requires that we assume that they have a (quasi-)stationary state.

For audio analysis tasks, we are interested in not just these discrete values, but in the spectrum of the signal, which contains information about its frequency and its temporal variation. The spectrum is obtained by converting the signal into individual spectral components through the process of the Discrete Fourier Transform (DFT) - derived by the Fast Fourier Transform (FFT) - in order to represent its amplitude or power in different frequency bins. This procedure (and further processing) certainly involves many more steps which will not be discussed in this context; for more information, the reader is

referred to W. Smith (1999) and Eyben (2016).

2.2.2 Signal Representation

The spectrum is therefore a representation of the signal in the frequency domain, commonly represented with its magnitude (absolute size). A more meaningful representation that denotes the evolution of frequency contents over time is done on a two-dimensional scale, using spectrograms. In a spectrogram, time is represented on the horizontal axis, frequency on the vertical axis, and the intensity of each frequency at a given point in time is depicted using color or grayscale, where more intense colors indicate stronger presence of energy at a specific frequency and at a given time. Acoustic analysis involves studying the spectral components of the signal with various approaches, each depending on the task.

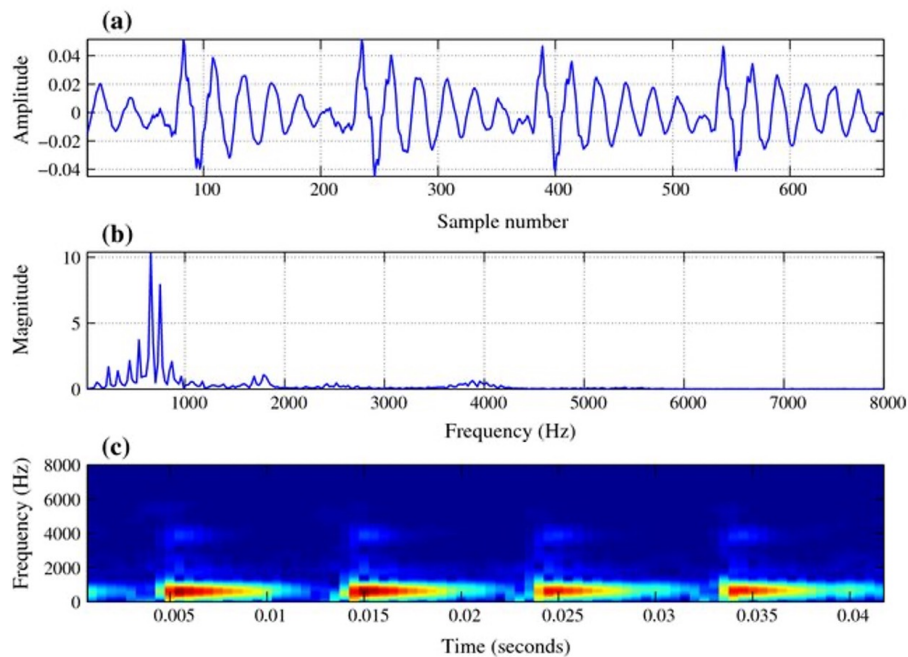


Figure 2.1: a) Plot of a speech signal, b) magnitude spectrum, c) spectrogram of speech signal representing the vowel /ae/.

2.3 State of the Art

2.3.1 Acoustic Biomarkers for detecting Mild Cognitive Impairment

In the past decade research efforts were focused on the automatic detection of individuals with progressed dementia symptoms, studying primarily the speech of patients diagnosed with Alzheimer's Disease, and the same sets of features were later evaluated for their efficacy in distinguishing earlier stages of cognitive decline with very promising results. Over time, innovative research accelerated along with advances in speech processing software and machine learning models. An important turning point was the ADReSS Challenge in 2020² which proposed shared tasks (AD classification and MMSME score regression), providing a forum for researchers to develop novel approaches using standardized datasets and evaluation metrics.

Overall, current research on the acoustic biomarkers for the detection of neurocognitive disorders revolves around two primary objectives:

- choosing the most accurate methods for quantifying these speech attributes, and
- finding novel features and reliable combinations for optimizing the classification performance of automatic screening tools.

Speech patterns suggesting cognitive impairment at the acoustic level are targeted by various acoustic features in the digital speech processing domain, which can be classified into three broad categories: prosodic, spectral and voice quality features - adopting the terminology of De La Fuente Garcia et al. (2020) and Eyben (2016).

This chapter will present the acoustic features studied as digital linguistic biomarkers of cognitive impairment, and their significance across studies that have achieved good performance in the automatic detection of dementia and MCI from speech.

²<https://luzs.gitlab.io/adress/>

2.3.2 Prosodic features

Prosody is a category that reflects suprasegmental features of speech, namely intonation, rhythm, and stress (Boschi et al., 2017). Its accurate representation in the digital domain is an intricate task, as the human auditory perception of prosody is formed with a complex combination of different attributes. When using acoustic features alone, it is typically measured in terms of speech rate, voice intensity, variations in frequency of the signal and formant structure, which are used as discriminative markers of cognitive impairment (Gabani et al., 2009).

2.3.2.1 Speech and silence duration

The first category of biomarkers associated with cognitive decline are temporal aspects of speech, measured with the quantity and duration of speech and silence segments. These can suggest various linguistic deficits caused by cognitive impairment; for example, higher and longer hesitations/pauses may indicate difficulty in articulation, deficits in syntax, lexical access, and discourse planning (Boschi et al., 2017). The ratio between speech and silence segments is also indicative of speech rhythm, especially when combined with lexical information.

Statistics on speech and silence segments have been essential features in many studies in the past years, which generally observed longer pauses and lower speech rate (percentage of speech found in the whole utterance) to be associated with dementia and Mild Cognitive Impairment (Roark et al., 2011; Satt et al., 2013; Beltrami et al., 2018). Similarly, studies have observed that speech of AD individuals contains voice segments that are higher in amount but smaller in length, in both spontaneous speech and reading tasks, compared to asymptomatic older adults (G. Meilán et al., 2013; König et al., 2015; Jarrold et al., 2014).

These features remain an integral part of speech applications that have been successful in distinguishing MCI patients from healthy subjects across different languages, often reported among the most significant variables. Specifically, an early study by König et al. (2015) used temporal acoustic features that also included the ratio of pauses to detect MCI and AD from healthy controls (with an accuracy of 79% and 87%, respectively), while another study in 2020 by Meilán et al. found that temporal acoustic features along with several frequency and intensity features differentiate people with MCI that develop Alzheimer from those

that do not.

2.3.2.2 Voice Activity Detection

The extraction of these temporal features relies primarily on the use of Voice Activity Detectors, which are the basis for most speech modelling tasks. The task of a voice activity detector is to discern speech samples from the signal from other noise-like sounds. Usually, each frame is assigned a probability value related to the likelihood that it contains human speech, or it is classified directly (as speech or non-speech) based on an assigned probability threshold. This probability is estimated by processing the signal information for calculating various features (such as short-time-energy, pitch detection, and spectral analysis) and by considering the present noise in the signal, which can of course vary constantly depending on the recording conditions (Ramirez et al., 2007).

A typical symptom of cognitive impairment is the deteriorating ability to access semantic content, which can lead to a higher amount of interlocutional pauses compared to healthy people of the same age group. Therefore, it is essential to use algorithms which can accurately detect smaller silence segments. The latest tools available are deep learning models that require training on large data, which is not practical for this task, given the limited corpora and their restrictions that would impose offline training. Many studies use in-house algorithms, but the majority do not explicitly mention what approach or tool was used; throughout the literature review the only exceptions identified were Calzà et al. (2021), who used a Spectral Subtraction VAD³ developed for interview speech, Nevler et al. (2017), who implemented the Speech Activity Detector from the Linguistic Data Consortium⁴ (dependent on the HTK corpus), and Ossewaarde et al. (2019), who used an algorithm proposed by Ramirez et al., 2004.

Additionally, hesitation in speech is not limited to silence regions, but also filled pauses - vocalizations such as “*uhm*”, “*ehm*” etc. Currently, VAD models are not sensitive enough to such pauses since they primarily detect voiced regions in a signal. For this reason and to achieve a more accurate segmentation overall, studies rely on text aligners for the reliable identification of speech and silence segments inside the sample.

³<https://github.com/enmwamak/ssvad>

⁴<https://github.com/Linguistic-Data-Consortium/ldc-bpcsad>

2.3.2.3 Energy/Loudness

Vocal intensity has been often studied as a distinguishing biomarker of cognitive impairment, on the account that the voice of cognitively impaired persons is often weaker and flatter in terms of prosody. A common and straightforward way to measure involves squaring the signal samples and calculating their average amplitude (RMS energy), to define short-term energy values; this acoustic feature has been implemented often in studies (Lopez de Ipiña 2012, Calzà et al., 2021), but is often reported less discriminative compared to other acoustic features. Loudness is also based on the signal's amplitude, but is measured on a logarithmic (decibel) scale, which reflects the perceptual intensity of speech and its changes, thus it is often implemented (Eyben., 2016; Haider et al., 2020; Martinc et al., 2021).

2.3.2.4 Pitch (F0)

Pitch is related to the perceived frequency of a voice, based on the number of vibrations per second produced by the vocal folds. It is one of the most fundamental elements of prosody, since it delivers emotion (for example, low pitch levels are related to somber emotions, like sadness) or intonation (i.e. adapting the tone of voice for stressing words or forming questions).

Pitch has been widely studied as a linguistic biomarker of cognitive impairment across the majority of studies, due to the fact that the voice of persons with cognitive impairment is often described as relatively monotone or “dry”; this perception however cannot be assessed directly, since pitch is composed of multiple frequencies produced with the vibration of the vocal folds. What we perceive as the tone of a voice, is a complex combination of fundamental frequency and its harmonics, which are multiples of the fundamental frequency appearing as additional peaks in the generated sound spectrum, contributing to our final interpretation (along with other elements which will be discussed further).

The generally accepted method is to rely on the height of fundamental frequency (F0), the primary frequency that is produced with vocal-fold vibration during phonation, and pitch detection algorithms analyze its structure and its harmonics over time for approximating the pitch contour, which better captures how humans interpret the signal in terms of periodicity.

Pitch features (mostly its minimum, maximum, mean and standard deviation) have been implemented across many studies that develop applications to distinguish dementia through speech, but often there are no associations with MCI found among the studies reviewed in this work. This may suggest the need to implement alternative measures of fundamental frequency, or f_0 -based approach, which is the approach of studies that focus on quantifying affective (emotional) prosody in MCI persons (Haider et al., 2020, Oh et al., 2023).

Furthermore, it is crucial to use appropriate measurement tools, in order to avoid the influence of individual speakers or a given utterance, which today can be a more straightforward task thanks to widely available standardized feature sets (Eyben, 2016). Nevertheless, estimating fundamental frequency alone is not enough to provide meaningful information for the overall spectrum, or provide concrete information to effectively model features of the human voice.

2.3.3 Spectral Features

2.3.3.1 Spectral descriptors and differences

To this end, there are a few basic measures implemented that represent either specific components of the spectrum, or its dynamics. Spectral descriptors are applied to single spectra, usually within a frame, describing elements about its energy content, its centroid (the center of gravity of the spectrum, which relates to what is perceived as “brightness” of the voice). Spectral moments and differences are also used as static descriptors of short-time spectra or estimates of spectral variations over time, respectively. Some of them have been reported to be significantly different between individuals with early or progressed cognitive impairment from healthy controls. In detail, subjects with MCI or dementia exhibit a higher spectral centroid and lower skewness, indicating an asymmetry towards higher frequencies when speaking (Fraser et al., 2016, Lopez de Ipiña et al., 2015, Calzà et al., 2021).

2.3.3.2 Formants

A formant is a concentration of acoustic energy around a particular frequency of a speech waveform. Besides the fundamental frequency of speech (F_0) there are several formants - enumerated as F_1 , F_2 , F_3 - each corresponding to a different frequency, roughly one within

each band of 1,000Hz. Formants and their variations determine the acoustic quality of different vowels, and each of them is connected with different resonances of the vocal tract-unlike fundamental frequency and harmonics, which are produced at the vocal folds. During speech production, sound undergoes modifications as it passes through the vocal tract, with formants effectively filtering - enhancing or attenuating - the original frequencies (for more information on source-filter theory, the reader is referred to Stevens, K.N., 2005). Like F0 and harmonics, formants can be seen in wideband spectrograms (Figure 2.2.)

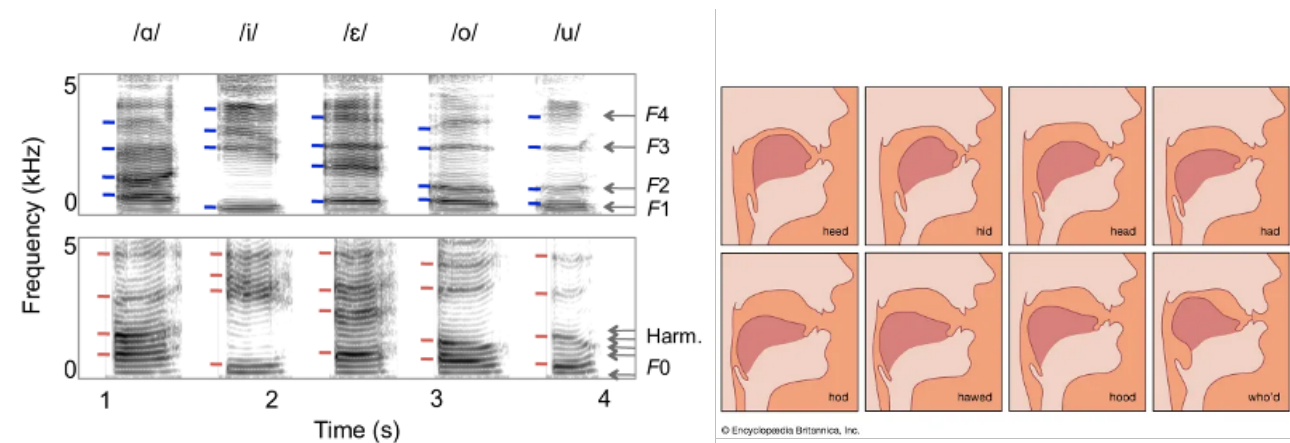


Figure 2.2: Spectrogram and vocal tract resonance at different vowels (Phonetics - Vowel Formants, Acoustics, Articulation | Britannica, n.d.)

In the context of investigating how neurocognitive disorders can affect speech, the incorporation of vowel formants in acoustic feature sets have given promising results: Themistocleous et al (2018) used a feature set with formants (F1 to F5), fundamental frequency and vowel duration to distinguish patients with MCI from healthy controls with 83% mean cross-validated accuracy, with a DNN model trained on a small dataset. In another study, they were assessed as predicting risk factors of cognitive deterioration in MCI patients, and it was suggested they can also uncover difficulties in the control of the sublaryngeal and laryngeal systems (Martinez-Nicolas et al., 2022), while Meilán et al. (2020) showed that particularly the standard deviation and the bandwidth size of the third formant (F3 and B3) were significantly different between AD from non-AD individuals.

2.3.3.3 Mel-Frequency Cepstral Coefficients

At this point, it is relevant to stress the importance of using appropriate methods for deriving meaningful information from acoustic signals, as improper analysis tools can introduce bias in cases where computed features are erroneously influenced by the unique subject's fundamental frequency, especially when measuring formants (Harrison 2013, Serrano et al., 2023.). A more comprehensive pitch tracking requires separating the influences of the source (the vocal-fold vibrations that produce sound) from the filter (the vocal tract). To this end, digital signal processing employs various techniques according to the type of information one needs from the signal (for more information, see Eyben, 2016). One of the most common methods -and very relevant to the scope of this study- is cepstral analysis, and in particular, its implementation for extracting Mel-Frequency Coefficients (MFCCs).

Mel-Frequency Cepstral Coefficients (MFCCs) are derived from a perceptual scale of pitches (the Mel-Frequency Scale) designed to mimic the human acoustic perception to changes in frequency. Human perception of sound is non-linear, which means that while changes in the lower frequency ranges are easily perceived, that ability progressively becomes more challenging in higher frequencies. The Mel scale is logarithmic and emphasizes lower frequencies more, thus reflecting our acoustic perception.

There are different types of cepstral coefficients, which are calculated as the discrete cosine transform logarithms of spectral power. This process involves applying the Discrete Fourier Transform to the logarithmic form of the spectrum's magnitude values, which effectively separates the source of the signal (i.e., the energy of the lungs) from the 'filter' of the signal (i.e., the upper vocal tract, which differentiates phonation output). From that process the signal can be represented in the cepstral domain, where the values of the source frequencies and vocal tract information are in distinct locations.

Mel-Frequency Coefficients specifically are derived during cepstral analysis with the application of overlapping, triangular-shaped filters which mimic the human auditory perception (Mel-Filterbanks) to the spectrum's magnitude. They represent the magnitude of specific frequency bands in the cepstral domain (the spectrum of a spectrum), where pitch periodicity can be more accurately represented (Young et al. 2006). In this way are very meaningful for speech analysis, particularly for reflecting the spectral envelope and

formant quality, by emphasizing dominant features of the speech spectrum and removing details that may be less important.

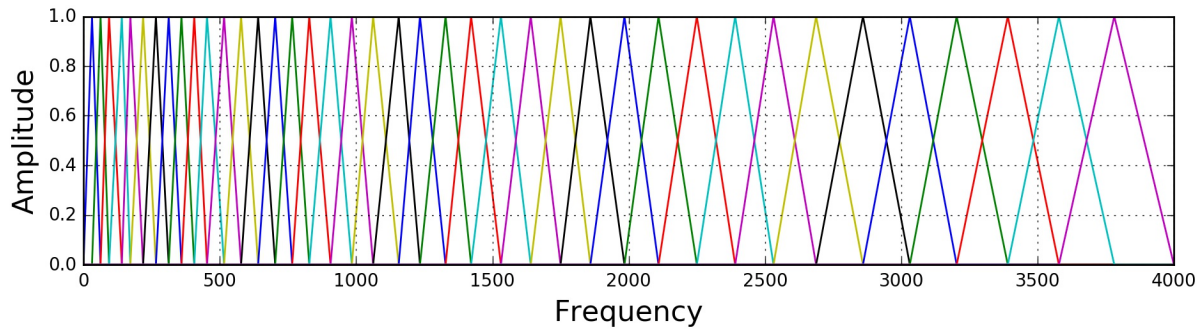


Figure 2.3: Mel-Filterbanks across frequency spectrum.

Currently, MFCCs are used in various speech modelling tasks - indicatively, for automatic speech recognition (Young et al. 2006), Computational Paralinguistics (Schuller, 2013). MFCCs are becoming widely used in works that study pathological markers of speech, since they offer important perceptual information and their calculations do not necessitate complex algorithms.

MFCCs can also be associated with articulatory traits, and studies have observed significant differences in their values between patients with cognitive impairment and healthy controls (Yamada et al., 2021; Jang H. et al., 2021). Usually, the range of the coefficients is used for calculating statistics regarding pitch periodicity, for the classification of both early and progressed stages of cognitive impairment (Fraser et al., 2015, Martinez et al., 2017, Egas-López et al., 2019; Calzà et al., 2021). More recently, a study used cepstral and voice quality features with the addition of Gammatone Cepstral Coefficients, which are calculated based on a more precise model of human auditory structure (Kumar et al., 2022); however, these are not widely used in relevant research, as they have a higher computational cost compared to MFCCs.

2.3.4 Voice Quality Features

In contrast to features that are traditionally associated to prosody (duration, energy, loudness, and pitch) which describe the overall melody and intensity of a speech, voice quality features describe the micro-prosodic variations within short-time frames (Schuller, 2013).

Voice quality biomarkers are generally used for reflecting laryngeal functioning during speech production. Normal phonation requires the combination of neuromotor, respiratory and laryngeal control, which can be affected by neurodegenerative diseases, including those leading to dementia (Martínez-Nicolás et al., 2021). Although changes in laryngeal function do occur as part of the ageing process, permanent and significant alterations in voice quality can indicate the initial symptoms of a degenerative disease (Woo et al., 1992). These changes typically manifest as hoarseness, breathiness and roughness in the voice (generally called dysphonia). Traditionally, voice quality is assessed by measuring the amount of Shimmer, Jitter, and Harmonics-to-Noise Ratio (HNR), which estimate irregularities in frequency, amplitude, and the amount of additive noise in the signal, respectively.

2.3.4.1 Jitter

Jitter measures the cycle-to-cycle differences of the fundamental frequency by calculating the percentage of variation in the time between consecutive F0 periods. A greater percentage of jitter will mean a higher pitch irregularity in sustained phonation and suggests a reduced control of the vocal-folds vibrations (Texeira et al., 2013).

2.3.4.2 Shimmer

The shimmer index refers to period-to-period amplitude variation in the voice signal. It quantifies the small intervals of instability, and is calculated as the percentage of amplitude fluctuation from one period (peak) to another. A higher shimmer value is associated with less stability and control of the sublaryngeal/pulmonary pressure.

2.3.4.3 Harmonics to Noise Ratio

The Harmonics to Noise Ratio (HNR), is the ratio of the energy of harmonic signal components (indicating speech) to the energy of the noise like components in a signal.

More noise in the signal can suggest more air produced along with speech, indicating greater effort in phonation.

These features are also leveraged for estimating tremulous voice, manifested as rapid interruptions of variations of frequency which are commonly associated with voice affected by neurodegenerative processes. Studies employing voice quality measures in their feature sets report great performance in detection of AD speech from healthy controls (Meilán et al., 2013; Lopez-de-Ipiña et al., 2015; Warnita et al., 2018) but also for patients with Mild Cognitive Impairment (Martinez et al., 2017, Haider et al., 2020, Themistocleous et al., 2020), while a study in new Spanish/Catalan cohort reported high diagnostic classification accuracy between controls and MCI with voice quality performing best out of all features (He et al., 2023). Voice quality features were investigated along with prosodic and spectral features in a longitudinal study by Lin et al. (2020), which reported that jitter and shimmer were found as the most significant features - with confidence interval up to 85% - associated with onset dementia in patients who were asymptomatic at the time.

Themistocleous et al. (2020) also investigated voice quality alterations in cognitive impaired subjects, by measuring differences within amplitudes of harmonics in speech and their relation to formant frequencies. They reported that MCI differed significantly from healthy subjects in H1-A3 - the amplitude of the first harmonic and the amplitude of the third formant (H1-A3)- similarly to a previous study between AD patients and healthy controls (Tanaka et al., 2017).

An advantage of voice quality features is that they do not rely on linguistic data, which can be more meaningful when investigating prosodic features, and can therefore be applied across different languages. Moreover, they may provide a “balance” against the declining accuracy of automatic transcription systems reported with elderly persons, which is precisely attributed to higher amounts of jitter and signal noise (Vipperla et al., 2008).

2.3.5 Self-similarity – Fractal dimension

Lastly, an interesting linguistic biomarker of pathological speech that has been implemented in literature derives from mathematics. Self-similarity is a trait of most fractal systems, that can be observed when a detailed inspection of their contents reveals repeated patterns within its structure. The fractal dimension reflects the geometry of a dynamical system given its multidimensional phase-space, and the degree of self-similarity – or complexity – is quantified by measuring the active degrees of freedom of the assumed dynamical system (Pitsikalis and Maragos, 2009, Maragos and Potamianos, 1999).

Fractals have been used in various frameworks for speech and bio-signals, but their application for identifying cognitive impairment through speech was leveraged by a series of studies by Lopez de Ipiña et al. (2013, 2015, 2018), suggesting that fractal dimensions can be a more thorough approach for quantifying speech irregularities and roughness, since a non-linear analysis can be more sensitive to subtle fluctuations. Indeed, in these studies and later in Calzà et al. (2021), the complexity of the patients' speech samples was significantly higher compared to the controls, and the implementation of this feature led to great performance even with different classifiers.

2.4 Overview

As technology improves, the study of digital linguistic biomarkers as potential early screening tools for cognitive impairment has flourished, with new research continuously contributing to the field and advancing speech processing software providing efficient and more accurate methods for detecting discrepancies in speech patterns. These ventures are granted to advance even more, given the constant breakthroughs of artificial intelligence models.

2.5 Limitations

In any case, however, their realistic application requires accuracy that can be confirmed only with rigorous evaluation using standardized metrics. Performance greatly depends on the evaluation methods chosen, dataset characteristics and classification algorithms. Past studies did not always present a baseline against which their results can be compared, with

some reporting only model accuracy which cannot attest for an equal performance in a real setting testing on unseen data. To this scope, in 2020, The ADReSS Challenge proposed shared tasks along with standardized datasets and predefined metrics, thus promoting a more comprehensive comparison of different approaches for Alzheimer's classification.

For what concerns acoustic features, reviewing past studies highlights their potential in capturing central linguistic manifestations of cognitive decline in the digital domain, and providing meaningful data for building robust classifiers. While these developments are very promising, they also emphasize the need to enhance study validity and replicability, so that future research and technology can be viable in a clinical setting. Studies may use different processing tools which can influence results, while few document explicitly the feature extraction process or important choices made. For what concerns the significance of features across research, some studies do not provide reports on each feature's statistical importance, while those who do sometimes omit to justify their choices and assumptions for choosing a specific test.

Additionally, the majority of studies and available corpora are focused on the English language, even though English speakers are a suboptimal representation of other language groups in clinical speech processing applications (Garcia et al., 2023). With regard to this serious limitation, a systematic review by De La Fuente Garcia et al. (2020) underlined the need for more linguistically diverse studies, proposing however that a shift towards acoustic features can alternatively address the issues of limited generalizability and non-English data scarcity. This can be encouraged by the fact that while most comprehensive studies include combined sets of various linguistic variables, acoustic features often outperform textual data, and have the added benefit of not relying on transcriptions, which can be very time-consuming if conducted manually, or prone to errors when done automatically.

Lastly, and most importantly, although the primary scope of this domain is to provide avenues towards the early detection of the disease, a substantial amount of the reviewed works focus on optimizing the identification accuracy for AD speech. While it is crucial to have a substantial body of work utilizing various methods for identification, priority should be placed on detecting the initial signs of neurocognitive disorders to realistically contribute to their early diagnosis.

2.6 Aims of this work

Considering the existing literature and demands of this field, the novelties of this thesis can be summarized as the following:

- The analysis of Italian samples with an enhanced acoustic feature set with respect to previous work, complementing the endeavor of building robust automatic classifiers, and supporting the field's understanding of the initial manifestations of dementia, as well as its cross-linguistic applicability;
- the development of a custom software that includes all needed tasks for the extraction of digital acoustic biomarkers, including pre-processing, voice activity detection, feature extraction and statistical representation on two levels (segment and global), along with its transparent description presented in this work;
- The implementation of novel and widely studied acoustic features specifically for detecting the early signs of neurocognitive disorder, leveraging standardized toolkits that can ensure reliable measurements and meaningful comparisons across studies, furtherly supporting the field's clinical potential.

3. Methodology

3.1 Project structure and objectives

To evaluate the discriminative power of acoustic features reviewed in the existing literature, a feature extraction pipeline was devised for the analysis of the speech samples of two groups of subjects, healthy controls and individuals diagnosed with Mild Cognitive Impairment. Before developing the package, a clear definition of its objectives and requirements was established, in order to address the aims of this project. This mainly involved two tasks: the selection of the most meaningful acoustic variables to be studied, and the feature extraction process. The following chapter describes the feature selection process, the dataset utilized, and the feature extraction pipeline, along with the specific computational techniques chosen.

3.1.1 Feature selection criteria

The selection of the acoustic features was done according to the following criteria:

1. **their use in peer-reviewed studies** aiming towards the distinction of neurocognitive disorder through speech between healthy participants and individuals diagnosed with Mild Cognitive Impairment, or progressed stages of dementia, given that the majority of relevant research focuses on AD speech;
2. **documented significance across studies**, namely:
 - by means of statistical testing, provided that the motivations and/or assumptions for each test was reported;
 - by extrinsic evaluation, i.e. noting whether implementing these features resulted in higher classification accuracy.

This led to the selection of the following acoustic feature categories : pitch (F_0), energy/intensity, spectral features (including Mel-Frequency Cepstral Coefficients), voice quality features (shimmer, jitter and harmonics to noise ratio), and the Higuchi Fractal Dimension, along with their statistical derivatives, which will be discussed thoroughly at the feature extraction process (Section 3.3.4).

3.2 Data Description

This project uses the dataset created in previous works on the automatic classification of mild cognitive impairment through digital linguistic biomarkers for Italian presented in Beltrami et al. (2018) and Calzà et al. (2021), supported by the OPLON project (Opportunities for active and healthy LONgevity, Smart Cities, Ministero Università e Ricerca, SCN 00176), and approved by the Ethical Committee of Azienda Ospedaliera Reggio Emilia (n. 148 2013/0013438). Given the sensitive nature of the data and restrictions imposed by the Italian legislation, it is not publicly available. The following section describes the dataset as it is documented in the aforementioned works.

3.2.1 Participants

The corpus is comprised of semi-spontaneous speech samples obtained from 92 Italian participants, 44 healthy controls (control group, CG) and 48 subjects with cognitive decline (pathological group, PG); all participants gave informed and written consent. The samples are balanced by sex, age (range 50-75) and education (primary school with great intellectual stimulation throughout the life span or junior high school; high school; academic degree).

The PG group had individuals from two outpatient clinical services for neurocognitive disorder evaluation and care. It is comprised of the following categories:

- **Mild Cognitive Impairment (MCI):** patients diagnosed with cognitive impairment that has not greatly impacted every day activities, divided in two subgroups to provide a balanced representation among clinical MCI types:
 - **amnesic MCI single domain (MCI-a):** patients who experience only a deficit in memory (16 subjects);
 - **multiple domain MCI (MCI-md):** patients who experience impairment in two or more cognitive domains.

- **Early Dementia (e-D):** patients who experience cognitive deficits that partially interfere with daily life (16 subjects).

	Healthy Controls	MCI Subjects	eD Subjects
Inclusion Criteria	MMSE ≥ 24 MoCA ≥ 18 No neurological pathologies, sensory impairment, or intellectual disability No problem in activities of daily living No familiarity with dementia	MMSE ≥ 18 No problem in activities of daily living No familiarity with dementia	MMSE ≥ 18 Need support for one or more activities of daily living No familiarity with dementia
Age	(61.60 $\pm$ 6.93) years	(64.34 $\pm$ 7.33) years	(66.38 $\pm$ 6.70) years
Education	(13.00 $\pm$ 3.92) years	(11.28 $\pm$ 4.35) years	(9.38 $\pm$ 4.01) years

Table 3.1: Inclusion Criteria, Age, and Education for Healthy Controls, MCI Subjects, and eD Subjects. MMSE: Mini-Mental State Examination, MoCA: Montreal Cognitive Assessment. Age and education are presented as mean $\pm$ standard deviation.

All participants completed an anamnestic interview which included anagraphic data, details regarding occupation or retirement, information about children, familiarity with neurodegenerative pathologies, clinical history, and pharmacotherapy. Additionally, each subject underwent a thorough neuropsychological evaluation which included a clinical interview, an assessment of cognitive, emotional, and behavioral features, an analysis of self-care abilities, an interview with a family member (when possible) and other medical examinations as part of the diagnostic work-up.

3.2.2 Data collection

For each participant, there are 3 recordings of semi-spontaneous speech, elicited by three mnemonic/linguistic tasks:

- **Task "Figure":** Picture description task (the picture depicted a living room with characters engaged in various actions);
- **Task "Work":** Description of a typical working day;
- **Task "Dream":** Description of the last dream the subject remembered.

These tasks involve different cognitive skills for prompting the semi-spontaneous linguistic production in the subjects, each tailored to semantic, autobiographical and episodic memory respectively, although all three require essential skills in recalling working memory (of what is being said), episodic memory (what has already been said) and prospective memory (planning what will be said).

Since the primary focus of this work is the extraction of acoustic features that can distinguish individuals with Mild Cognitive Impairment from healthy controls, the final analysis and results are focused on the discrimination between the control and MCI groups, while the entire corpus is used to assess how the relevance of each feature in the acoustic set might change using more samples.

3.3 Data Processing

After reviewing the past works to select the features associated with MCI through speech, an acoustic analysis package was built in python to process the data and return the final features. The following section describes the overview of the pipeline and the steps taken to extract each feature.

3.3.1 Pipeline Overview

The processing of the speech samples is co-ordinated by the class `AudioProcessor` of the package, which takes as input the audio recording (in a wave format) and handles the entire processing of the sample, from pre-processing, speech and silence identification (by calling the `VADProcessor`) to the extraction of features. Functions used throughout the process, i.e. reading the signal content and sampling rate of the file are defined in a dedicated module *utils.py*, which uses the corresponding functions from `audinterface` package.¹

Initially, an `AudioProcessor` object is created along with the attributes of the signal array and sampling frequency. Next, Voice Activity Detection (Section 3.3.3) is performed to identify speech and silence regions in the signal.

¹<https://audeering.github.io/audinterface/api/audinterface.utils>

In the pipeline, features relating to the speech and silence duration and quantity and the self-similarity of the waveform are extracted directly with the AudioProcessor, while the open-source framework openSMILE (Eyben, 2016) was used to extract pitch, spectral (including MFCCs) and voice-quality low-level descriptors (Section 3.3.4.2).

Two baseline feature sets available with the toolkit were used:

- **The INTERSPEECH 2016 ComparE set**, the latest baseline acoustic feature set available in the toolkit, developed for the Computational Paralinguistics Challenge (Schuller et al., 2016), for extracting pitch (F0), RMS energy, loudness, Mel-Frequency Cepstral Coefficients, spectral descriptors and voice quality features;
- **the Geneva Minimalistic Parameter Set (GeMAPS)** for extracting formant-related features, as well as fundamental frequency measured in semitones, which is deemed as a representation more robust against individual speaker differences.

These feature sets have been systematically evaluated on several databases for classification tasks using paralinguistic features of speech, and optimized over time to include more noise-robust feature computations from real-life conditions (Eyben, 2016, pp.239-241). Furthermore, they are used in many state-of-the-art works on cognitive impairment classification tasks using speech analysis, including the ADReSS challenge (Luz et al., 2020).

The information for each feature extracted is stored into pandas Dataframes. All the final features, besides the majority of the speech and silence statistics set which are calculated on the entire utterance, are retrieved on two levels (Section 3.3.5). Namely:

- **on a global/file level**, i.e. by averaging the frame values of the entire sample to obtain their mean and standard deviation,
- **on a segment level**, by first computing the mean and standard deviations for each feature of each speech segment of the sample, and then returning the final values for the entire sample considering the mean values of each segment.

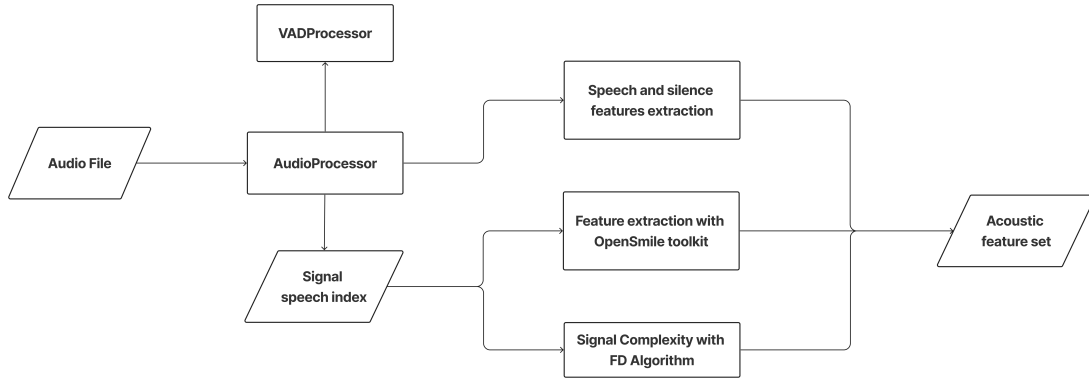


Figure 3.1: Pipeline Overview

3.3.2 Pre-processing

Acoustic analysis typically involves a series of processing steps on the waveform for a meaningful and accurate representation. In this pipeline, all features extracted, besides the speech and silence segments information and the fractal dimension (performed on the time-domain signal), are implemented using the openSMILE package (Eyben, 2016) which handles the necessary initial processing of the samples for acoustic analysis, including scaling the audio samples to the range $[-1, +1]$, except for the computation of the MFCC features in which case the audio is scaled the 16-bit sample range of $[-32767, +32768]$ (Eyben, 2016, p. 260).

Hence, the pre-processing tasks involve the initial resampling of the audio files from the original sampling frequency of 44.1 kHz to that of 16 kHz, which is typically used for speech analysis tasks. This step is handled once an AudioProcessor class object is initialised, which takes as input the audio file to be processed and an input sampling frequency (if specified) by the user. The resampled data is stored in a new folder inside the directory of the input file and is accessible from the AudioProcessor for being passed as a parameter to the Voice Activity Detection tool and the feature extraction functions.

3.3.3 Voice Activity Detection

The first task regards identifying speech and silence segments of the signal, which is handled by the class VADProcessor of the package. This step is required at the beginning in order to obtain the speech segments to be analysed, and for extracting the features relating to speech and silence. The following sections describe the Voice Activity Detection algorithm chosen

for this project, its evaluation, and its implementation inside the pipeline.

3.3.3.1 SS-VAD Algorithm

Following Calzà et al. (2020), this project used the SS-VAD by Mak & Yu, an energy-based voice activity detector developed for interview speech, that performs spectral subtraction to reduce background noise before identifying speech segments in the signal (Yu & Mak, 2011). The choice of this algorithm lies in the fact that it developed specifically for processing interview speech files of the NIST (US National Institute of Standards and Technology) Speaker Recognition Evaluation² series, files which contain various issues that can hinder accurate speech detection, as described by the authors:

1. extremely low Signal to Noise Ratio (SNR), i.e., a large amount of noise energy;
2. periodic background signals that interfere with the speech signal's content;
3. impulsive signals caused by plosive sounds or the speaker speaking too close to the microphone.

These are issues that can typically arise during interview recordings, despite precautions taken to ensure recording quality. Thus, SS-VAD employs spectral subtraction to efficiently increase the signal-to-noise ratio in the speech regions, while decreasing the signal's amplitude in the non-speech (background) regions, to prevent non-speech sounds being classified as speech.

SS-VAD Algorithm Description

1. First, the average noise frequency spectrum is estimated.
2. For each frame (multiplied by a hamming window):
 - (a) its SNR is calculated, and
 - (b) the background noise spectrum is subtracted from the spectrum of the noisy signal's frame; if there is a particularly low SNR, a constant, low noise is set to the frame.

²<https://sre.nist.gov/>

3. The denoised speech signal is reconstructed using the inverse Fourier Transform and dehamming window.
4. In the denoised speech signal, the background frames are identified by searching for the frames with the lowest amplitude, and the peak frames by searching for the frames with the largest amplitude.
5. Then, a detection threshold is determined; here the influence of signal spikes in the decision can cause low-energy speech frames to be mistakenly detected as non-speech. Therefore, the threshold is estimated using a linear combination of the minimum of the signal peaks and the mean of background amplitudes.
6. Frames are classified as speech if their smoothed amplitude is higher than the obtained threshold; otherwise, they are determined as silence regions.

The SS-VAD approach is documented extensively in the papers by Mak & Yu (2011; 2014), and the source code (written in C) can be found at its Github repository.³

3.3.3.2 Voice Activity Detection Evaluation

Since the corpus used for the analysis in this study did not include annotations regarding speech segment timestamps, the SS-VAD evaluation was done using speech samples from the CLIPS corpus⁴, which contains multiple sets of speech recordings in Italian. The segmentations of the VAD were checked with the manual transcriptions of 10 samples of around one to two minutes each from the *radiotelevisivo* subset, which were recorded in various noise conditions.

The labelled data were obtained by parsing the transcriptions of each file - written in XML format - and the prediction set was created after calling the VAD for each sample and retrieving the start and end times of speech and silence segments. In the context of this work, the correct estimation of both speech and silence segments is equally important, so the metrics were implemented for assessing both speech and silence identification performance.

³<https://github.com/enmwmak/ssvad>

⁴<http://www.clips.unina.it/it/corpus.jsp>

The information of the labels and predictions was stored in two types of dictionaries that contained binary arrays for each key-segment (1 for speech, 0 for silence, and vice-versa) to allow an easier evaluation.

An important issue at this step was that the provided transcriptions did not include smaller pauses but only significantly long silence segments or hesitations, and other datasets that were consulted followed the same logic.⁵ The choice to continue with the samples from the CLIPS corpus was based on the fact that they resembled the conditions of actual data used for evaluating the acoustic features, namely longer speech segments, different Signal to Noise Ratio across files, and a range of different voices of Italian persons, particularly elderly voices.

3.3.3.3 Evaluation metrics

To get a comprehensive evaluation, several metrics were employed: firstly, accuracy and F1 score (computed as the harmonic mean of precision and recall), which are basic standardized metrics for evaluating Natural Language Processing applications, and additionally, the NIST defined Detection Cost Function (DCF), specifically for evaluating Voice Activity Detection Algorithms (Joglekar & Hansen, 2022). These measures and their calculations are described below.

Accuracy:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Recall:

$$Recall = \frac{TP}{TP + FN}$$

Precision:

$$Precision = \frac{TP}{TP + FP}$$

F1 Score:

$$F1 = 2 \times \frac{Precision * Recall}{Precision + Recall}$$

Where:

- **TP (True Positives)** is the number of correctly predicted positive instances, or correct

⁵Among which the *Librivox* dataset, available at <https://www.openslr.org/94/>, and the more recent *LaMIT* corpus, by Di Benedetto et al., 2022.

detections;

- **TN (True Negatives)** is the number of correctly predicted negative instances, or correct rejections;
- **FP (False Positives)** is the number of incorrectly predicted positive instances, or insertions;
- **FN (False Negatives)** is the number of incorrectly predicted negative instances, or deletions.

These values were calculated by comparing the prediction and label sets, implementing a prediction tolerance value of 0.2 seconds (Dinkel et al., 2020; Kraljevski et al., 2015).

Moreover, the **Detection Cost Function (DCF)** was included, defined as:

$$\text{DCF} = (1 - \gamma) \cdot P_{\text{miss}} + \gamma \cdot P_{\text{fa}},$$

where false alarm rate P_{fa} is the percentage of non-speech labels being misclassified as speech, and miss rate P_{miss} is the percentage of speech labels being misclassified as non-speech, and the weighting factor $\gamma = 0.25$ penalises the missed positives more heavily (Joglekar & Hansen, 2022). The evaluation is documented at the *Notebooks* folder of the github repository.⁶

3.3.3.4 VAD Performance

The F1 score (Figure 3.2) suggests an overall good performance for speech detection, which is also supported by the low Detection Cost Function. However, there is a considerably smaller precision (and a lower F1 score) in the identification of silence regions, which could be attributed to the fact that the VAD recognizes smaller silence segments than documented in the manual annotations.

3.3.3.5 VAD implementation in the pipeline

Using the SS-VAD software, speech and silence segments are identified within the VADProcessor class of the pipeline. Voice Activity Detection is performed by executing the compiled binary file which performs the processes described above. Since the pipeline is

⁶https://github.com/chloepdpd/MasterThesisDHDK/blob/main/Notebooks/VAD_EVAL.ipynb

Metric	Speech	Silence
Accuracy	0.830	0.830
Precision	0.962	0.515
Recall	0.821	0.908
F1 Score	0.879	0.620
DCF	0.115	0.055

Table 3.2: VAD performance for speech and silence detection with a 0.2 s tolerance

written in Python, the SS-VAD binary is executed using the *subprocess* module⁷, a native Python package that allows the execution of system commands and external programs written in different languages within Python scripts.

The SS-VAD program was built for processing NIST files which are in sphere format, therefore a function that converts the samples accordingly using the *sox* package is included in the class. The VADProcessor also handles some further processing which aims to remove small silences that are not considered significant and are part of normal locution; the threshold implemented was that of 500 ms.

The final output returned is a Pandas DataFrame which contains speech and silence segments, their start and end times (in sample positions and in milliseconds), duration of each segment and total segment duration.

This information is stored by the AudioProcessor for the following tasks:

1. calculating the temporal features regarding speech and silence,
2. using the content of denoised signal for the features computed directly on the waveform, and
3. creating an index of the speech segments in the form of a pandas DataFrame denoting the segment id and its timestamps in milliseconds, which is used for extracting the features computed from speech regions only.

⁷<https://docs.python.org/3/library/subprocess>

3.3.4 Feature Extraction

3.3.4.1 Speech and silence duration

The first group of features are extracted by calculating information from the speech and silence duration of the samples. This set of variables has proven to be among the most significant distinguishing qualities of cognitive impairment, therefore multiple measures were taken into account after evaluating their relevance in previous studies.

For each sample, the following statistics were calculated based on the speech and silence segments:

- **Mean, median, standard deviation of speech and silence segments** (Satt et al., 2011; Calzà et al., 2021).
- **Transformed Phonation Rate** (Singh et al., 2001; Roark et al., 2011; Calzà et al., 2021). This feature is calculated as the arcsin of the square root of the Phonation Rate (PR): $\arcsin(\sqrt{PR})$, where the Phonation Rate is the ratio of Total Phonation Time (the total speech time without pauses) to the Total Locution Time (total speech time including pauses): $PR = TPT / TLT$.

The arcsin transformation provides a normally distributed measure within each participant group: $\arcsin(\sqrt{PR})$.

- **Mean ratio of silence to speech**, calculated by dividing the mean silence duration by the mean speech duration of each segment (König et al., 2015).
- **Mean number of silences** per utterance, specifying the average number of silences produced by a speaker in one second of speech. It is calculated by dividing the total number of silences by the total locution time or TPT (Demiroglu et al., 2015).

For this task, there is a dedicated function of the AudioProcessor class, which retrieves the duration arrays for each type (speech and segment) and uses the *numpy* package for deriving statistics (mean, median and standard deviation) and the angular (arcsin) transformation. The final values were stored in dictionaries and returned in a pandas DataFrame, which is eventually merged with the rest of the features for each sample, based

SPE_mean	Mean speech duration (ms)
SPE_median	Median speech duration (ms)
SPE_std	Standard Deviation of speech duration (ms)
SIL_mean	Mean silence duration (ms)
SIL_median	Median silence duration (ms)
SIL_std	Standard Deviation of silence duration (ms)
SIL_count	Average number of silences per utterance
SIL_to_SPE_mean	Mean ratio of silence to speech duration
T_phon_rate	Transformed Phonation Rate.

Table 3.3: Features calculated from speech and silence duration. Pauses are considered as silence segments above 0.5 seconds.

on the file index.

3.3.4.2 Features implemented with openSMILE

The openSMILE package was used to extract pitch, spectral (including MFCCs) and voice-quality low-level descriptors, computed on frames of 20ms and a step of 10ms, while the fundamental frequency related features are measured with frame size of 60 ms (and step of 10 ms). Each feature is smoothed with a symmetric moving average window of 3 frames to produce the final value for each frame.

To obtain the features, each sample was processed according to its signal index, obtained with the AudioProcessor after calling the VADProcessor class, that indicates the id of the speech segments and their timestamps in milliseconds in the form of a pandas Timedelta object, so that it conforms to the openSMILE toolkit interface.⁸

Pitch (F_0). Pitch is typically measured based on fundamental frequency (F_0), which measures the rate of the vibrations of the vocal folds during speech production, and is expressed in Hz or cycles per second. We consider the mean F_0 and standard deviation (Martínez-Sánchez et al., 2012; Lopez-de-Ipiña et al., 2013; Calzà et al., 2021), measured both linearly and in semitones, available the ComparE and GeMAPS set of the openSMILE package, respectively. The addition of the semitone version was made since expressing fundamental frequency in semitones is suggested to mitigate gender differences as

⁸<https://audeering.github.io/audinterface/api/audinterface>

suggested in Gomez-Moreira et al. (2018) and a recent systematic review by De La Fuente-Garcia et al. (2021). Both versions were integrated in order to compare their relevance.

The default pitch detection algorithm utilized in the openSMILE toolkit is based on the Subharmonic Summation (SHS) method by Dik J. Hermes (1998). This involves a series of steps, including the use of the signal's harmonic structure to identify the correct pitch, spectral peak enhancement, auditory weighting, and the summation of different spectrum representations. Lastly, post-smoothing is applied using dynamic programming (Viterbi algorithm) to determine the optimal trajectory of pitch values and associated voicing probabilities (Eyben, 2016, p.65-72).

Energy and Loudness. To measure signal intensity in voiced regions, two features were extracted:

- **Root Mean Square Energy:** $RMS = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}$, where N is the number of samples, and x_i refers to each individual sample of the signal at the i -th position;
- **Loudness (dB)**, obtained from a Mel-band power spectrum (described in detail below), and weighted by an equal loudness curve function for the frequency dependent loudness sensitivity of the human auditory system (Eyben, 2016, pp. 21-21, pp. 56-57).

As before, these two features are often reported in previous as markers of voice intensity, thus it was deemed useful to integrate both of them for assessing their efficacy in capturing potential speaker differences.

Spectral Features. To obtain a comprehensive representation the distribution of energy across different frequency components, several **spectral descriptors** were computed. Firstly, the **spectral centroid**, computed as the center of gravity of the spectrum:

$$S_{centroid} = \frac{\sum_{m=1}^M f(m) \cdot X(m)}{\sum_{m=1}^M X(m)},$$

where M is the number of frequency bins, $f(m)$ is the frequency of the m -th bin, and $X(m)$ is the magnitude of the m -th bin in the spectrum; this measure can refer to the perceptual

"brightness" of a sound, and is found to be higher in individuals with cognitive impairment, indicating that the energy of the signal is concentrated at higher frequencies (Lopez de Ipiña 2015; Calzà et al., 2021).

Besides the spectral centroid, a non-linear psychoacoustic version included in the ComparE feature set was employed. It is defined as **perceptual sharpness**, proposed by Zwicker and Fastl (1999), which computes the spectral centroid scaled on a Bark frequency scale, on which the distances correspond with perceptually equal distances of loudness, corresponding to the first 24 critical bands of hearing. (Eyben., 2016, p.39-41).

Additionally, the spectrum's second to fourth **spectral moments** are considered: spectral variance, skewness and kurtosis, respectively. The spectral variance (also called spectral spread), measures the spread of spectral energy around the spectral centroid, with spectral skewness measuring the asymmetry of that distribution towards higher or lower frequencies (resulting to a lower or higher skewness), while spectral kurtosis indicates the peakedness of the distribution (Eyben, 2016, pp.39-40). In order to compute these statistical moments, the spectrum $X(m)$ must be converted to a Probability Mass Function (PMF), which provides the probability distribution of spectral energy at a particular frequency bin:

$$p_X(m) = \frac{X(m)}{\sum_{i=m_l}^{m_u} X(i)}.$$

This way, the full spectral range is considered, from $m_l = 1$ to $m_u = M$, with $X(m)$ as the power spectrum, i.e., $X(m) = X_P(m)$. Therefore, spectral variance S_{variance} is defined as:

$$S_{\text{variance}} = S_{\sigma}^2 = \sum_{m=m_l}^{m_u} (F(m) - S_{\text{centroid}})^2 p_X(m),$$

and the spectral standard deviation S_{σ} as:

$$S_{\sigma} = \sqrt{S_{\text{variance}}},$$

spectral skewness S_{skewness} as:

$$S_{\text{skewness}} = \frac{1}{S_{\sigma}^3} \sum_{m=m_l}^{m_u} (F(m) - S_{\text{centroid}})^3 p_X(m),$$

and spectral kurtosis S_{kurtosis} as:

$$S_{\text{kurtosis}} = \frac{1}{S_{\sigma}^4} \sum_{m=m_l}^{m_u} (F(m) - S_{\text{centroid}})^4 p_X(m).$$

Lastly, a dynamic descriptor, **spectral flux**, which reflects changes in spectral shape. It is defined as the quadratic version of the signal's spectral difference, calculating the distance between two successive spectra:

$$S_{\text{flux}}^{(k)} = \sum_{m=m_l}^{m_u} \left(X^{(k)}(m) - X^{(k-1)}(m) \right)^2,$$

where each short-time spectrum $X^{(k)}(m)$ at discrete time (frame index) k is treated as a point X in an M -dimensional space (Eyben, 2016, p.42). While there exist additional measures for describing spectral attributes (both static and temporal), this specific set was chosen as it has been leveraged in previous works as voice markers of cognitive decline (Fraser et al., 2016, Meilán et al., 2020).

Mel-Frequency Cepstral Coefficients. Features deriving from the signal's spectrum also involve the **Mel-Frequency Cepstral Coefficients**, which reflect the energy variations between frequency bands of a signal, effectively representing overall spectral characteristics of voice. Typically, the first 13 14 coefficients are used - often excluding the 0th coefficient, MFCC-0, which is computed from the total energy of the spectrum - as they refer to the lower range frequencies of the vocal tract that yield the most meaningful information (Hernández-Domínguez et al., 2018), and have been very effective at identifying AD patients in previous literature (Satt et al., 2014; Fraser et al., 2018; de la Fuente Garcia et al., 2020; Meghanani and Ramakrishnan, 2021).

Mel-Frequency Coefficients are derived with cepstral analysis applying overlapping triangular shaped filters (Mel-Filterbanks) to the frequency spectrum's magnitude (Figure 4.3). These filters follow the Mel-Frequency Scale, a conceptual scale of pitches (Stevens et al., 1937; Stevens and Volkman, 1940) based on the human auditory perception, where perceptual pitch periodicity can be more accurately represented. In the openSMILE

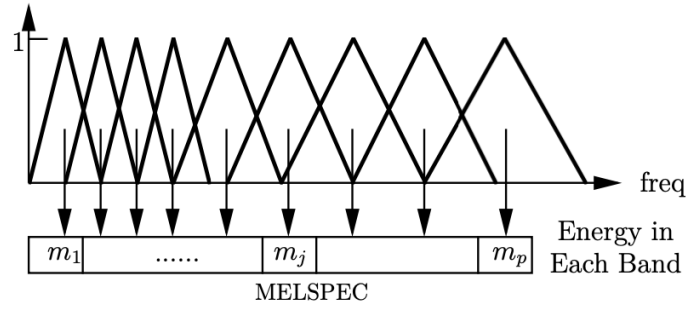


Figure 3.2: Mel-frequency scale filterbanks on the log-magnitude spectrum, as in Young et al., 2001.

package, MFCCs are by default computed from a power Mel-spectrum (Eyben, 2016, pp.28-32, 60-61), which is formed from the linear scale magnitude spectrum using triangular Mel-Scale filters that have 50 % overlap. In detail, the filterbank of B filters with centres evenly spaced between $f_{\min}$ and $f_{\max}$ is created as follows: A constant bandwidth β for the filters on the target scale is assumed:

$$\beta = \frac{1}{B+1} \left(f_{\max}^{(\text{scale})} - f_{\min}^{(\text{scale})} \right).$$

The centre frequencies $f_c^{(\text{scale})}(b)$ on the target scale for B filters $b = 1 \dots B$ and the lower and upper bounds ($b = 0$ and $b = B + 1$) are computed like so:

$$f_c^{(\text{scale})}(b) = f_{\min}^{(\text{scale})} + b\beta.$$

The power spectrum shape of each filter is then given by:

$$\Phi_b(m) = \begin{cases} 0 & \text{for } f^{(\text{scale})}(m) \leq f_c^{(\text{scale})}(b-1) \\ \frac{f^{(\text{scale})}(m) - f_c^{(\text{scale})}(b-1)}{f_c^{(\text{scale})}(b) - f_c^{(\text{scale})}(b-1)} & \text{for } f_c^{(\text{scale})}(b-1) < f^{(\text{scale})}(m) \leq f_c^{(\text{scale})}(b) \\ \frac{f_c^{(\text{scale})}(b+1) - f^{(\text{scale})}(m)}{f_c^{(\text{scale})}(b+1) - f_c^{(\text{scale})}(b)} & \text{for } f_c^{(\text{scale})}(b) < f^{(\text{scale})}(m) \leq f_c^{(\text{scale})}(b+1) \\ 0 & \text{for } f^{(\text{scale})}(m) > f_c^{(\text{scale})}(b+1) \end{cases},$$

with

$$f^{(\text{scale})}(m) = \Theta_{\text{scale}} \left(F^{(\text{lin})}(m) \right),$$

where Θ_{scale} refers to the Mel-Frequency scale. The filterbank is typically designed for the range from 20 Hz, for avoiding the bias of DC - direct current components, which pertain to the average value of the signal (Smith, 1999, pp.13-14; Eyben, 2016, p.16)- to 8,000 Hz. Next, the logarithm is applied to $X_p^{(mel)}(b)$:

$$X_p^{(log,mel)}(b) = \begin{cases} \log(X_p^{(mel)}(b)) & \text{if } X_p^{(mel)}(b) < X_{\text{floor}} \\ X_{\text{floor}} & \text{otherwise} \end{cases} .$$

The value X_{floor} is used to floor very small values that result from quantisation noise during the calculation, in order to avoid large negative log values. Then, on the log Mel-spectrum $X_p^{(log,mel)}(b)$, applying Discrete Cosine Transformation results in K Mel-cepstral coefficients $C^{(mel)}(k)$, where K in our case is 14:

$$C^{(mel)}(k) = \sqrt{\frac{2}{B}} \sum_{b=0}^{B-1} X_p^{(log,mel)}(b) \cos\left(\frac{\pi k}{B} \left(b + \frac{1}{2}\right)\right) .$$

Lastly, the Cepstral coefficients $C^{(mel)}(k)$ are filtered in a way to emphasize lower order coefficients (liftering). Given a liftering coefficient L , the liftering applies weighting with a sine function:

$$C^{(mel)'}(k) = C^{(mel)}(k) \left(1 + \frac{L}{2} \sin \frac{\pi k}{L}\right) .$$

For this project, **mean and standard deviation** (of the speech frames smoothed with symmetric moving average window of 3 frames) of **MFCCs 1-14** were considered using the ComparE feature set, computed from a power Mel-spectrum described above, with range of 20–8000 Hz and cepstral liftering of $L = 22$ (Eyben, 2016, pp. 28-32, 60-61).

Formants. In pathological voice, the range of articulatory movements decreases, which can result in altered formant frequencies in subjects affected by cognitive degeneration (Vizza et al., 2018). Following previous studies (Meilán et al., 2018; Themistocleous et al., 2020), the mean and standard deviation of the central frequency and bandwidth of the first three formants (F1-F3) were extracted.

In the GeMAPS set of the openSMILE toolkit, these are computed employing LPC

(Linear Predictive Coding) analysis, which is based on the source-filter model of speech and predicts the values of a signal as a linear function of previous samples (Makhoul, 1975; Schuller, 2013). This gives an approximation $\hat{x}(n)$ of the original signal $x(n)$, constructed by a weighted sum of p previous samples (Schuller 2013):

$$\hat{x}(n) = - \sum_{i=1}^p a_i x(n-i),$$

where p is the number of previous samples considered (also referred to as the order of the linear predictor) and the weighting coefficients a_i are the linear predictor coefficients; the inversion in front of the sum simplifies the equations for the computation of the linear predictor coefficients.

Formants can also be estimated directly from the magnitude spectrum, however modelling the spectral envelope using Linear Prediction is typically preferred since it can more effectively separate the influences of the fundamental frequency in voiced sounds. In the magnitude spectrum, the range of possible F_0 values overlaps with the range of the first formant F_1 , and especially in female voices, spectral peaks of F_0 and F_1 may not be easily separable which can lead to inaccuracies (Eyben, 2016, pp.51-53).

The LP spectrum has a sufficiently high resolution that allows formant frequencies to be estimated by identifying spectral peaks, and counteracts the possible influence of the F_0 , since in its spectral envelope the influence of F_0 has been completely removed by the linear predictor, if the length of the linear predictor $p \ll N_{F_0}^{(T)}$ ($N_{F_0}^{(T)}$ being the length of the fundamental frequency period in samples). The configuration for formant extraction in the openSMILE toolkit using LPC involves a series of steps, along with error minimisation functions; thus, for a comprehensive description the reader is referred to Eyben, 2016, p.48-55.

Voice Quality Features. The voice quality set of features includes Jitter, Shimmer and Harmonics to Noise Ratio, which are typical measures used throughout research for assessing voice quality in ageing voices, and more recently, voice pathology attributed to neurocognitive disorder (Martinez et al., 2017, Haider et al., 2020, Themistocleous et al., 2020). As in previous studies, we consider the mean and standard deviation of these features, as well as the delta Jitter which measures frame-to-frame variance, computed

using the ComparE set of the openSMILE toolkit (Eyben, pp. 73-78, pp. 129-130). In detail:

Jitter, the period to period variation of fundamental frequency, is measured as the absolute difference between consecutive pitch periods. With the length of the first period $n' - 1$ as $T_0(n' - 1)$ and the length of the second period n' as $T_0(n')$, period to period Jitter is defined as:

$$J_{pp}(n') = |T_0(n') - T_0(n' - 1)| \quad \text{for } n' > 1,$$

while its variance across frames (delta period-to-period Jitter) is defined as the difference between two consecutive J_{pp} values:

$$J_{ddp}(n') = |J_{pp}(n') - J_{pp}(n' - 1)| \quad \text{for } n' > 2.$$

Shimmer measures the variation between consecutive periods of the signal's amplitude, therefore with the peak to peak amplitude as $A(n') = x_{\max, n'} - x_{\min, n'}$ the local Shimmer (period to period) is given as:

$$S_{pp}(n') = |A(n') - A(n' - 1)|$$

.

Harmonics-to-noise ratio (HNR) describes the relationship between the energy present in the harmonic region (associated with voice) and in the non-harmonic region, indicating noise-like sounds. It is a logarithmic measure typically computed from the average waveform:

$$HNR_{\text{wf}, \log} = 20 \cdot \log_{10} \left(\frac{E_{\text{harm}}}{E_{\text{noise}}} \right) \text{ dB}$$

.

3.3.4.3 Self-similarity - Higuchi Fractal Dimension

The last feature computed regards the self-similarity or complexity of the signal, using the Higuchi Fractal Dimension (Lopez de Ipiña et al., 2013; 2015; 2018; Calzà et al., 2021). The Higuchi Fractal Dimension (Higuchi, 1988) measures the fractal dimension of a time series by dividing it into discrete sequences, much like the concept of signal framing.

In detail, given a time series $X : \{1, \dots, N\} \rightarrow \mathbb{R}$ consisting of N data points and a parameter $k_{\max} \geq 2$, the Higuchi Fractal dimension (HFD) of X is calculated in the following way: For each $k \in \{1, \dots, k_{\max}\}$ and $m \in \{1, \dots, k\}$, define the length $L_m(k)$ by

$$L_m(k) = \frac{N-1}{\lfloor \frac{N-m}{k} \rfloor k^2} \sum_{i=1}^{\lfloor \frac{N-m}{k} \rfloor} |X_N(m+ik) - X_N(m+(i-1)k)|.$$

The average length of all the series, $L(k)$, is defined by the average value of the k lengths $L_1(k), \dots, L_k(k)$:

$$L(k) = \frac{1}{k} \sum_{m=1}^k L_m(k).$$

The slope of the best-fitting linear function through the data points $\{(\log(1/k), \log L(k))\}$ is defined to be the Higuchi fractal dimension of the time-series X .

In our case, the time series is the speech signal, while the k parameter refers to the maximum interval length between points, which is important since it defines the observation scale by which we measure self-similarity. The fractal dimension algorithm iterates over various subseries lengths (up to the number k) to assess the fractal properties of the signal.

To compute the self-similarity of the speech samples, the pipeline includes a dedicated function which applies the HFD algorithm, importing the open-source *HiguchiFractalDimension* package for python, which can be used either directly or as a python wrapper after the compilation of the C library that contains the HFD algorithm.⁹

⁹<https://github.com/inuritdino/HiguchiFractalDimension>

For faster execution, this C library is compiled at the project's initialization using the subprocess module, and the imported module *hfd.py* uses it to perform the fractal dimension analysis.

As before, we only consider the speech segments of the samples, but since we are working with the values of the time-domain waveform, the AudioProcessor contains additional processes for obtaining the correct input for the HFD algorithm. These include obtaining the path of the denoised signal corresponding to the sample being processed, which, after the denoising process of the VAD, is saved in the directory of the original input file. Next, the signal values which correspond to speech regions only are retrieved, according to the start and end times of the segments in sample numbers, which are also part of the speech index used for the previous tasks. This results in two representations of the time-series signal: one signal array, containing all the speech regions values, and an array of signal segments each corresponding to a unique speech segment of the original file. To calculate values on a file level (over the entire utterance), the complete signal array is passed to the HFD function, while for a segment-level representation, the function processes each speech segment separately.

Furthermore, three different k parameters were implemented, with length of 360, 640, and 1280 samples, respectively. A short window will give a highly local estimation which will be more sensitive to changes in the waveform, while with a longer window, some details may be lost but the fractal dimension better anticipates the overall characteristics of the signal (Lopez-de-Ipiña et al., 2013).

3.3.5 Final processing

The last step of the pipeline involves averaging the low-level descriptors obtained from the speech samples, in order to obtain a meaningful comparison across groups, and for providing the final data eventually processed as feature vectors for classification tasks.

This is handled by the *features* module of the package, which is used after the audio file processing is complete. For the features computed with the openSMILE toolkit, the summary metrics are calculated from the DataFrame returned during the feature extraction, which contains the frame by frame values. The global statistics are computed by averaging the frame values to obtain their mean and standard deviation, while the segment

level statistics are returned by first computing the mean and standard deviations for each feature on independent speech segments, and then performing a new averaging function to the values of each segment, thus returning the final values for the utterance.

This process concerns solely the features obtained by the opensmile toolkit, since the rest of the features' statistics - i.e. speech and silence duration (Section 3.3.4.1) and the self-similarity index (Section 3.3.4.3) - are handled directly inside their corresponding extraction functions; the computation of these features in particular does not return frame-level values, thus it would not be possible to calculate the statistics across the entirety of the features.

All the final features are then combined into a unique DataFrame - one for each summary metric level - based on the file name index. Thus, when the extraction process is complete, the features of the processed file can be returned on both levels (global and segment) each stored in a corresponding DataFrame, where they are organised into columns, along with the index of the file. For extracting features of multiple files, the *features* module of the package contains a function (*process_folder*) that takes as input a directory and performs the acoustic analysis for each file included that has a wave format. Thus, in this case, the final output is a DataFrame which is indexed according to each file name, and each column contains the values of the extracted features. An example of the output can be seen at the Github repository of the project.¹⁰

¹⁰<https://github.com/chloeppd/MasterThesisDHDK/blob/main/Notebooks/demo.ipynb>

Step	Description	Class
1.	Resample input file to the input sampling rate and call the VADProcessor.	AudioProcessor (Section 3.3.2)
2.	Perform denoising and VAD on the resampled file and return a Dataframe depicting speech and silence segments and their timestamps and its denoised version (file path).	VADProcessor (Section 3.3.3)
3.	Obtain speech index from the speech and silence segments; store the speech index and the denoised file path as attributes.	AudioProcessor (Section 3.3.3.5)
4.	Compute the features of speech and silence duration from the speech and silence segments DataFrame	(Section 3.3.4.1)
5.	Extract features from the segments containing speech, determined by the speech index passed to the openSMILE functions.	(Section 3.3.4.2)
6.	Compute the self-similarity value for each file, applying the HFD algorithm on the denoised version of the signal, on the speech segments determined by the speech index in sample numbers. Return one value of the whole signal (file-level calculation) and the mean and standard deviation of the values of each speech segment (segment-level).	(Section 3.3.4.3)
7.	Calculate summary statistics (file and segment level).	+ features.py
8.	Combine into one Dataframe for each level and save in .csv format.	(Section 3.3.5)

Table 3.4: Summary of steps.

4. Analysis

4.1 Statistical analysis

After finalizing the feature extraction package, the speech samples of the participants were processed for extracting the selected acoustic biomarkers. The function *process_folder* was used to automatically process all files and save the obtained features for each group.

In order to assess the discriminative power of the extracted features, the Kolmogorov-Smirnov (KS) test was applied to obtain each feature's significance across the two groups, following the methodology of the previous studies that designed the corpus (Beltrami et al., 2018; Calzà et al., 2021), given the limited amount of data. This test is used for comparing the distributions of continuous variables between two samples, assuming they are not normally distributed (non-parametric test), with the null hypothesis that both groups were sampled from the same distribution. The KS test was applied for comparing the values of each feature between the two groups using the *scipy.stats* package.¹

4.2 Results

For a comprehensive analysis of the results, this section is comprised of three parts. The first part of the analysis refers to relevance of each feature in distinguishing the participants with MCI from the healthy controls of the corpus, which is the primary aim of this study. Additionally, the contribution of each speech elicitation task is documented, in order to their usefulness in capturing speech disfluencies and alterations in the MCI group. Lastly, we observe the distributions of the analysis results when comparing the values of the control subjects (CON) to the entirety of the pathological group (PG), hence taking into account the subset of the corpus comprised of participants diagnosed with early dementia (e-D). This

¹two sample test, with the default null hypothesis that the group samples are identical; available at <https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.kstest.html>

particular comparison is performed as it may allow a more concise understanding into the usefulness of each feature in monitoring disease progression.

In this chapter, the means and standard deviations of the normalized values of the two groups are presented for each feature, along with their statistical significance. The *p-value* represents the probability that the cumulative values of the two groups are identical, as evaluated by the Kolmogorov-Smirnov (KS) test. When applicable, each feature is marked according to its observed significance.²

4.2.1 Evaluation of acoustic features as biomarkers for MCI identification

4.2.1.1 Prosodic features

Table 4.1 outlines the significance of the first set of prosodic features, namely the variables computed from the speech and silence regions of the participants' samples: statistics of speech and silence duration, transformed phonation rate, average silence count and mean ratio of silence duration to speech duration (whose calculation is documented in Section 3.3.4.1).

On average, the speech of individuals with MCI considered in this study contains shorter speech segments than the healthy subjects group, and a higher amount of pauses that are longer in duration. This is furtherly supported by the significant differences in the Transformed Phonation Rate, the mean ratio of the silence and speech durations, and the average silence count, which all highlight the prevalence of more hesitations compared to the healthy group. Besides the median value of the silence segments, all the measures of this set are highly discriminative markers (Table 4.1).

With regard to the next set of prosodic features, we can observe a pattern of lower mean values of intensity for the MCI group, with solely the mean RMS energy being statistically relevant with a probability value of 0.024 when calculated from the entire speech sample, and of 0.013 at the segment level (Table 4.2). No significant alterations are observed in RMS energy variability or any the loudness features. For what concerns the measures of **pitch**

²Namely: *** for $p < 0.001$, ** for $p < 0.01$, and * for $p < 0.05$.

Speech & silence	CON_mean	CON_std	MCI_mean	MCI_std	p-value	Signif.
SPE_ms_mean	0.221	1.216	-0.303	0.446	<0.001	***
SPE_ms_median	0.174	1.249	-0.239	0.385	<0.001	***
SPE_ms_std	0.261	1.184	-0.359	0.492	<0.001	***
T_phon_rate	0.322	0.817	-0.442	1.067	<0.001	***
T_phon_rate_segment	0.228	0.925	-0.313	1.024	0.001	***
SIL_ms_mean	-0.197	0.870	0.271	1.108	0.007	**
SIL_ms_median	-0.163	0.760	0.224	1.231	0.139	
SIL_ms_std	-0.185	0.979	0.254	0.983	<0.001	***
SIL_count	-0.309	0.835	0.424	1.060	<0.001	***
SIL_to_SPE_mean	-0.312	0.503	0.429	1.315	<0.001	***

Table 4.1: Speech and silence features; *T_phon_rate* refers to Transformed Phonation Rate, computed on both levels (entire utterance and on each segment).

mean and standard deviation values, while the MCI group displays elevated means of fundamental frequency and a lower variability at the segment level, there are no meaningful differences in any of the values between the two groups (Table 4.3).

File level	CON_mean	CON_std	MCI_mean	MCI_std	p-value	Signif.
RMSEnergy_mean	0.178	1.010	-0.244	0.943	0.024	*
Loudness3_mean	0.093	0.950	-0.128	1.061	0.089	
RMSEnergy_std	0.099	0.968	-0.136	1.037	0.235	
Loudness3_std	0.052	0.941	-0.072	1.082	0.433	
Segment level	CON_mean	CON_std	MCI_mean	MCI_std	p-value	Signif.
RMSEnergy_mean	0.178	1.009	-0.244	0.945	0.013	*
Loudness_std	0.105	0.946	-0.144	1.062	0.064	
Loudness_mean	0.092	0.947	-0.127	1.066	0.153	
RMSEnergy_std	0.137	0.976	-0.189	1.013	0.163	

Table 4.2: Intensity features. Means and standard deviations and significance at the KS test.

4.2.1.2 Spectral Features

The spectral features, on the other hand, show striking differences between the two groups. There are consistent patterns found in the ranges of the Mel-Frequency Cepstral Coefficients - low, middle, and higher - which demonstrate lower variations of energy at different frequency bands in the speech samples of the MCI group. In detail, the means of the first three coefficients are the most discriminative with probability values at less than 0.001, at both levels (Tables 4.8 and 4.9). Additionally, there are notable decreases in the

File level	CON_mean	CON_std	MCI_mean	MCI_std	P-Value	Signif.
F0semitone_mean	-0.179	1.013	0.111	0.939	0.122	
F0final_mean	-0.172	0.971	0.107	1.000	0.144	
F0final_std	-0.004	0.930	0.086	1.124	0.649	
F0semitone_std	0.027	1.031	-0.028	1.003	0.974	
Segment level	CON_mean	CON_std	MCI_mean	MCI_std	P-Value	Signif
F0semitone_std	0.144	0.984	-0.058	0.988	0.083	
F0_mean	-0.172	0.971	0.107	1.000	0.144	
F0semitone_mean	-0.176	1.013	0.119	0.952	0.158	
F0_std	-0.004	0.930	0.086	1.124	0.649	

Table 4.3: F0 (linear and semitone starting from 27.5Hz) mean and standard deviations.

standard deviations of MFCC 7-8 and 10 , as well as in the mean values of MFCC 2, 6 and 13. This tendency is evident at both levels, although some details are lost at the file level.

Similarly, the **spectral descriptors and moments** implemented are highly relevant (Table 4.4). In the speech samples of the MCI group, the two measures employed for quantifying voice brightness (spectral centroid and psychoacoustic sharpness) both indicate a consistent concentration of higher frequencies in the central mass of the spectrum, with no significant variations as suggested by their corresponding standard deviation values. Furthermore, these distributions show a significantly lower asymmetry (skewness), which reflects a disproportionate concentration of energy at higher tones, and a low kurtosis of the spectral peaks. The spectral variance and flux highlight a diminishing variability both at the local level (variance) and when estimated over time (flux). Again, this pattern is reflected at both levels but the values computed from the entire speech samples seem less precise.

Table 4.5 displays the comparison between the last set of spectral features extracted, namely the center frequencies and bandwidths of the first three formants. In the samples of the MCI participants there is a significantly lower standard deviation of the bandwidth of the second format of MCI subjects compared to the healthy controls, as well as a lower mean frequency at the third formant (F3). Additionally, we observe a higher variability at its frequency when computed from the global average, and a lower standard deviation at its bandwidth (F3_bandwidth_std) when computed from the segment values. The other formant features are not particularly discriminative as evaluated with the K-S test.

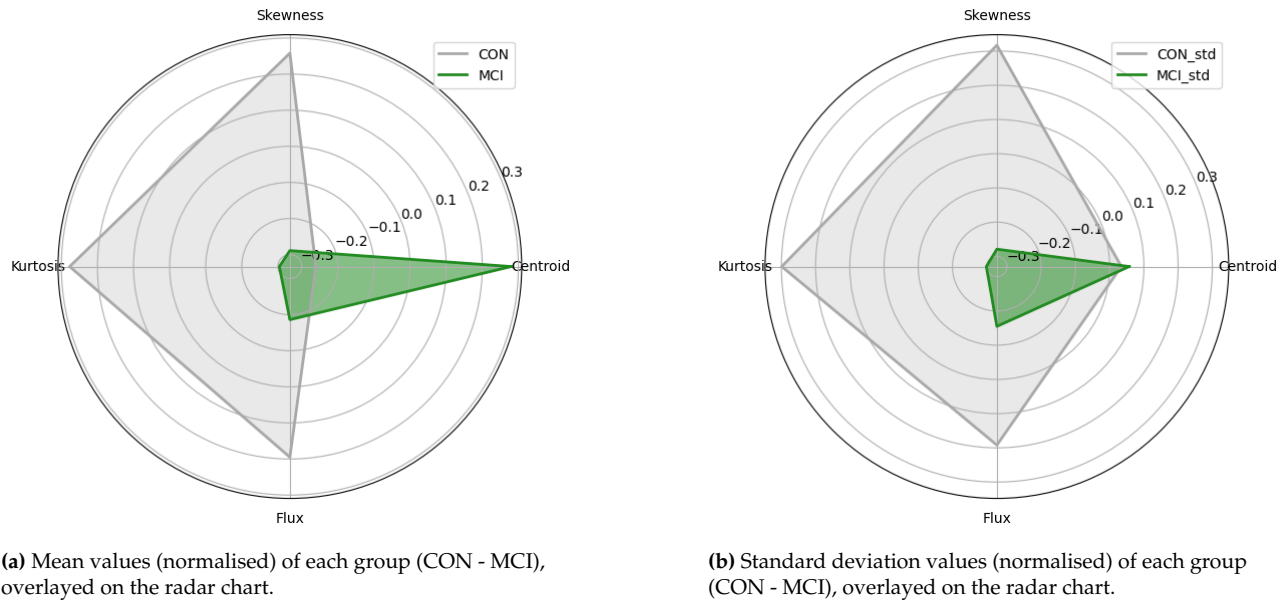


Figure 4.1: Radar charts depicting the spectral centroid, skewness, kurtosis, and flux values, each represented as a point in the axis. The length of each spoke represents the value of the corresponding variable; longer segments towards the perimeter denote higher values.

File level	CON_mean	CON_std	MCI_mean	MCI_std	P-Value	Signif.
psySharpness_mean	-0.287	0.935	0.309	1.019	0.005	**
spectralSkewness_std	0.256	1.101	-0.265	0.762	0.007	**
spectralKurtosis_std	0.272	1.149	-0.293	0.665	0.010	*
spectralKurtosis_mean	0.284	1.164	-0.319	0.607	0.010	*
spectralCentroid_mean	-0.270	0.913	0.309	1.063	0.014	*
spectralSkewness_mean	0.245	1.126	-0.279	0.713	0.048	*
spectralFlux_mean	0.188	0.977	-0.194	0.924	0.131	
psySharpness_std	-0.082	0.993	0.091	1.040	0.364	
spectralFlux_std	0.161	0.998	-0.136	0.903	0.488	
spectralCentroid_std	-0.092	0.982	0.106	1.051	0.568	
spectralVariance_std	-0.036	0.980	0.005	1.042	0.977	
Segment level	CON_mean	CON_std	MCI_mean	MCI_std	P-Value	Signif.
spectralCentroid_mean	-0.228	0.926	0.314	1.023	<0.001	***
psySharpness_mean	-0.243	0.940	0.334	0.993	<0.001	***
spectralSkewness_std	0.257	1.124	-0.353	0.663	<0.001	***
spectralSkewness_mean	0.229	1.128	-0.315	0.687	<0.001	***
spectralKurtosis_mean	0.242	1.157	-0.333	0.600	<0.001	***
spectralKurtosis_std	0.251	1.165	-0.345	0.566	0.001	***
spectralFlux_mean	0.165	1.001	-0.227	0.963	0.018	*
spectralVariance_mean	-0.171	0.796	0.235	1.196	0.024	*
spectralFlux_std	0.150	1.031	-0.206	0.928	0.072	
spectralVariance_std	0.052	1.067	-0.071	0.907	0.249	
psySharpness_std	-0.000	1.056	0.001	0.928	0.303	
spectralCentroid_std	-0.010	1.045	0.014	0.946	0.423	

Table 4.4: Spectral descriptors: means and standard deviations.

Global level F1-F3	CON_mean	CON_std	MCI_mean	MCI_std	p-value	Signif
F3frequency_mean	0.147	1.084	-0.202	0.841	0.009	**
F2bandwidth_std	0.160	0.992	-0.220	0.979	0.014	*
F3frequency_std	-0.136	0.895	0.187	1.111	0.033	*
F1frequency_mean	-0.113	1.026	0.155	0.952	0.051	
F1frequency_std	-0.110	1.013	0.152	0.972	0.168	
F2bandwidth_mean	-0.018	0.889	0.024	1.144	0.191	
F2frequency_mean	0.027	1.101	-0.037	0.853	0.242	
F3bandwidth_std	0.037	0.951	-0.051	1.072	0.271	
F1bandwidth_mean	-0.013	0.972	0.019	1.047	0.311	
F2frequency_std	-0.018	0.958	0.025	1.064	0.423	
F3bandwidth_mean	-0.037	1.083	0.051	0.882	0.601	
F1bandwidth_std	0.027	1.003	-0.038	1.005	0.697	
Segment level F1-F3	CON_mean	CON_std	MCI_mean	MCI_std	p-value	Signif.
F2bandwidth_std	0.157	0.969	-0.216	1.012	0.005	**
F3frequency_mean	0.164	1.077	-0.226	0.843	0.006	**
F3bandwidth_std	0.052	0.880	-0.071	1.150	0.044	*
F2frequency_mean	0.044	1.092	-0.061	0.866	0.064	
F2bandwidth_mean	-0.012	0.885	0.016	1.149	0.114	
F1frequency_mean	-0.084	0.998	0.116	1.002	0.130	
F1bandwidth_std	0.096	1.105	-0.132	0.827	0.130	
F1frequency_std	-0.077	1.017	0.106	0.977	0.144	
F3frequency_std	-0.104	0.925	0.143	1.088	0.328	
F3bandwidth_mean	-0.079	0.939	0.109	1.078	0.346	
F2frequency_std	0.015	0.964	-0.021	1.057	0.896	
F1bandwidth_mean	0.005	1.049	-0.007	0.939	0.984	

Table 4.5: Means and standard deviations of first three formants frequencies and bandwidths.

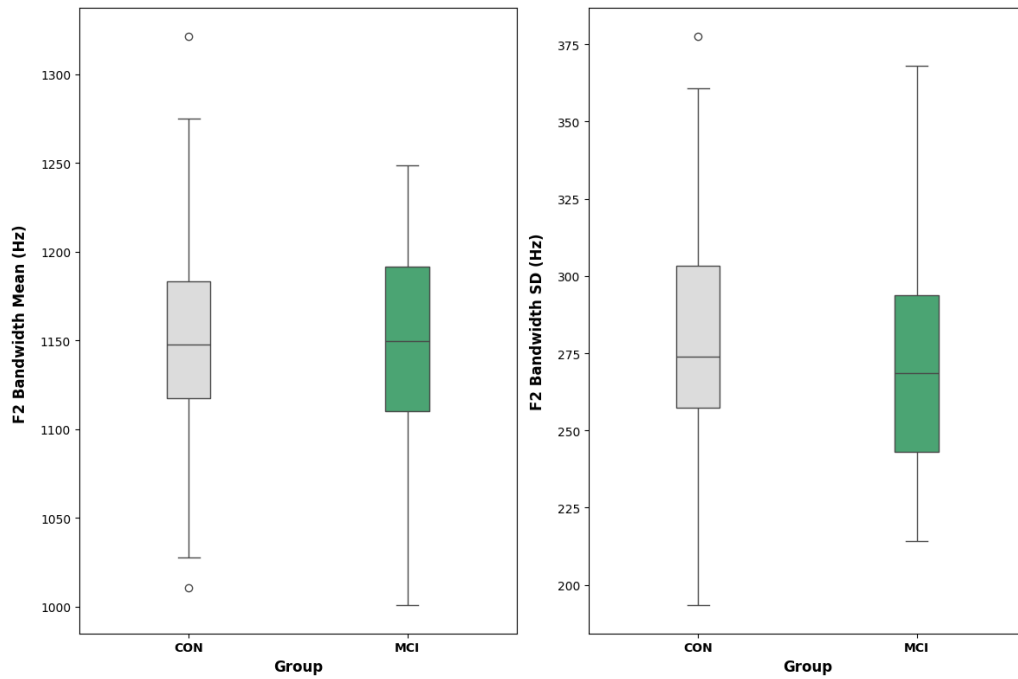


Figure 4.2: Box plots demonstrating the mean bandwidth of the second formant (left) and its standard deviation (right) between the two groups (CON - MCI).

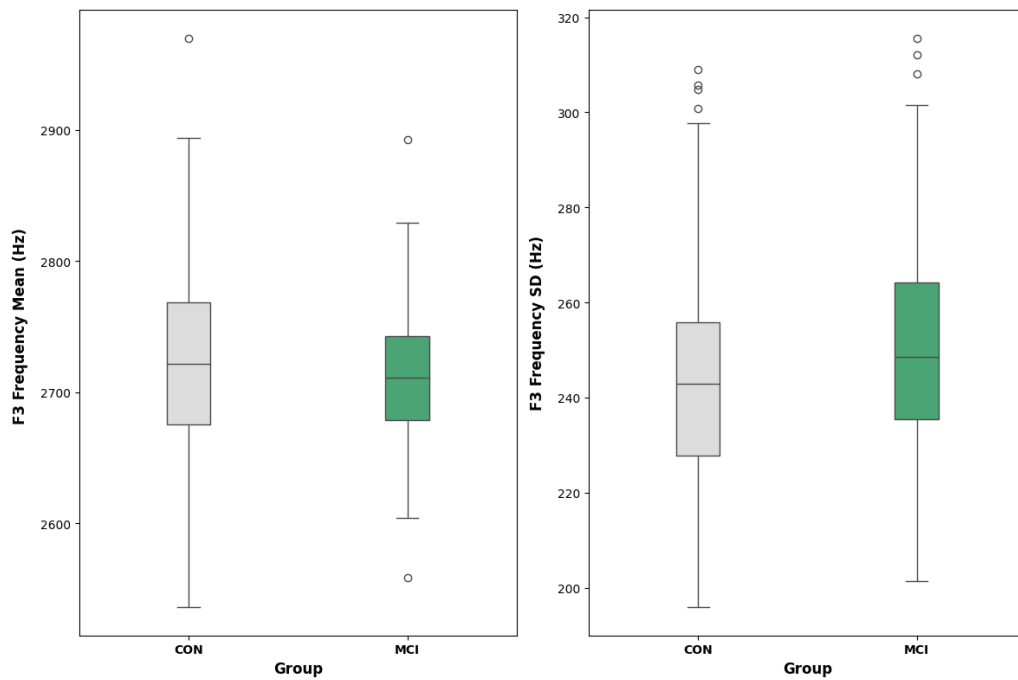


Figure 4.3: Box plots demonstrating the mean frequency of the third formant (left) and its standard deviation (right) between between the two groups (CON - MCI).

4.2.1.3 Voice Quality features

For what concerns voice quality measures, no significant differences exist between the two groups, with the exception of the local shimmer mean and the harmonics-to-noise ratio (mean and standard deviation) computed from the segments average (Table 4.6). These features indicate more amplitude perturbations during phonation, along with higher noise

components which vary compared to the healthy control group.

File level VQ	CON_mean	CON_std	MCI_mean	MCI_std	p-value	Signif.
shimmerLocal_mean	-0.102	0.947	0.140	1.062	0.122	
shimmerLocal_std	0.062	0.900	-0.085	1.127	0.126	
logHNR_mean	0.107	0.959	-0.147	1.046	0.235	
logHNR_std	0.021	0.978	-0.030	1.039	0.697	
jitterDDP_std	0.042	1.000	-0.058	1.008	0.732	
jitterLocal_std	0.035	1.002	-0.049	1.006	0.766	
jitterDDP_mean	-0.015	0.969	0.020	1.051	0.800	
jitterLocal_mean	-0.020	0.978	0.028	1.038	0.831	
Segment level VQ	CON_mean	CON_std	MCI_mean	MCI_std	p-value	Signif.
logHNR_std	0.109	0.964	-0.149	1.039	0.018	*
shimmerLocal_mean	-0.107	0.976	0.147	1.024	0.033	*
logHNR_mean	0.095	0.973	-0.130	1.032	0.041	*
jitterDDP_mean	-0.029	0.949	0.041	1.075	0.077	
jitterLocal_mean	-0.035	0.965	0.048	1.054	0.168	
shimmerLocal_std	0.086	0.948	-0.118	1.067	0.249	
jitterLocal_std	0.086	1.046	-0.118	0.932	0.364	
jitterDDP_std	0.088	1.051	-0.122	0.922	0.393	

Table 4.6: Voice quality features (global and segment level).

4.2.1.4 Self-similarity

The fractal dimension feature also contributes as a highly discriminative feature, with the MCI group showing particularly lower self similarity values, (i.e. higher complexity in the signal), especially when using a larger window (Table 4.7); at the segment level however, group differences are not significant.

File level HFD	CON_mean	CON_std	MCI_mean	MCI_std	p-value	Signif.
HFD_w360	0.140	0.932	-0.193	1.066	0.037	*
HFD_w640	0.211	0.875	-0.290	1.095	0.014	*
HFD_w1280	0.236	0.858	-0.324	1.096	0.003	**
Segment level HFD	CON_mean	CON_std	MCI_mean	MCI_std	p-value	Signif.
HFD_w360_std	0.040	1.176	-0.054	0.698	0.179	
HFD_w360_mean	-0.016	1.148	0.022	0.762	0.555	
HFD_w640_mean	-0.000	1.165	0.000	0.728	0.185	
HFD_w640_std	0.040	1.184	-0.054	0.680	0.144	
HFD_w1280_std	0.039	1.193	-0.053	0.658	0.099	
HFD_w1280_mean	0.004	1.183	-0.006	0.685	0.179	

Table 4.7: Signal self-similarity measured with the Higuchi Fractal Dimension (HFD) and window size in samples.

MFCC 1-14 (File level)	CON_mean	CON_std	MCI_mean	MCI_std	P-Value	Signif.
mfcc[3]_mean	0.340	1.006	-0.368	0.838	<0.001	***
mfcc[2]_mean	0.286	1.093	-0.316	0.735	<0.001	***
mfcc[1]_mean	0.254	0.920	-0.310	1.050	<0.001	***
mfcc[8]_std	0.120	0.959	-0.190	1.038	0.001	***
mfcc[3]_std	0.158	1.045	-0.219	0.953	0.002	**
mfcc[12]_mean	0.244	0.961	-0.248	0.952	0.002	**
mfcc[7]_std	0.198	0.992	-0.331	0.954	0.003	**
mfcc[10]_std	0.118	0.969	-0.180	1.077	0.005	**
mfcc[13]_mean	0.204	0.965	-0.214	0.862	0.025	*
mfcc[6]_mean	0.184	0.816	-0.078	1.096	0.026	*
mfcc[2]_std	0.169	1.114	-0.251	0.822	0.033	*
mfcc[5]_std	0.118	1.011	-0.186	1.014	0.069	
mfcc[4]_mean	0.129	1.051	-0.068	0.916	0.069	
mfcc[1]_std	0.119	0.984	-0.078	0.992	0.072	
mfcc[10]_mean	0.124	1.030	-0.151	0.966	0.118	
mfcc[4]_std	0.111	1.018	-0.148	0.992	0.130	
mfcc[5]_mean	0.160	0.933	-0.123	1.067	0.203	
mfcc[11]_mean	-0.070	1.021	0.154	0.969	0.235	
mfcc[9]_mean	0.013	1.017	0.022	1.008	0.303	
mfcc[7]_mean	-0.028	1.106	-0.021	0.856	0.328	
mfcc[14]_std	-0.091	0.880	0.091	1.151	0.374	
mfcc[12]_std	-0.038	0.921	0.088	1.135	0.433	
mfcc[6]_std	-0.027	0.892	-0.022	1.088	0.613	
mfcc[9]_std	-0.001	1.021	0.052	0.991	0.685	
mfcc[13]_std	0.001	0.923	-0.010	1.134	0.732	
mfcc[14]_mean	0.060	0.960	0.047	1.001	0.870	
mfcc[11]_std	-0.052	0.977	0.072	1.026	0.879	
mfcc[8]_mean	0.027	0.966	0.021	0.980	0.946	

Table 4.8: MFCC features on a file level

MFCC 1-14 (Segment level)	CON_mean	CON_std	MCI_mean	MCI_std	P-Value	Signif.
mfcc[2]_mean	0.296	1.099	-0.324	0.721	<0.001	***
mfcc[3]_mean	0.327	0.985	-0.366	0.877	<0.001	***
mfcc[1]_mean	0.259	0.938	-0.288	1.017	<0.001	***
mfcc[7]_std	0.290	0.932	-0.242	0.947	<0.001	***
mfcc[10]_std	0.202	0.901	-0.116	1.047	0.001	**
mfcc[3]_std	0.267	0.975	-0.186	0.890	0.003	**
mfcc[8]_std	0.203	0.900	-0.111	1.027	0.005	**
mfcc[6]_mean	0.178	0.819	-0.112	1.145	0.006	*
mfcc[2]_std	0.237	1.040	-0.175	0.849	0.009	*
mfcc[13]_mean	0.198	0.972	-0.188	0.838	0.017	*
mfcc[12]_mean	0.228	0.986	-0.227	0.911	0.018	*
mfcc[1]_std	0.240	0.910	-0.106	0.911	0.027	*
mfcc[10]_mean	0.129	1.048	-0.136	0.941	0.037	*
mfcc[5]_std	0.198	0.952	-0.139	1.014	0.042	*
mfcc[4]_mean	0.120	1.048	-0.079	0.942	0.046	*
mfcc[13]_std	0.110	0.808	0.021	1.148	0.062	
mfcc[4]_std	0.207	0.934	-0.112	0.990	0.067	
mfcc[6]_std	0.103	0.846	0.005	1.125	0.118	
mfcc[11]_std	0.093	0.861	0.067	1.063	0.191	
mfcc[11]_mean	-0.069	1.042	0.144	0.949	0.209	
mfcc[5]_mean	0.150	0.924	-0.136	1.092	0.222	
mfcc[9]_std	0.102	0.913	0.057	0.997	0.337	
mfcc[12]_std	0.072	0.836	0.087	1.105	0.355	
mfcc[14]_std	0.048	0.829	0.087	1.148	0.383	
mfcc[7]_mean	-0.015	1.098	-0.032	0.864	0.509	
mfcc[14]_mean	0.037	0.955	0.076	0.995	0.720	
mfcc[8]_mean	0.000	1.000	0.046	0.943	0.755	
mfcc[9]_mean	0.025	1.043	-0.000	0.970	0.927	

Table 4.9: MFCC features on a segment level

4.2.2 Evaluation of speech elicitation tasks

To obtain a more concise understanding of the relevance the semi-spontaneous speech tasks employed in the study (outlined in Section 3.2), the significance of the extracted acoustic features were evaluated across the samples obtained from each task utilized for interviewing the participants, namely the picture description ("**Picture**", Table 4.10), working day ("**Work**", Table 4.11), and dream narration ("**Dream**", Table 4.12). The tables display all the features that are ranked significant - with a p-value below 0.05 - at the Kolmogorov-Smirnov test. As in the previous section, the means and standard deviations of the normalized values of the two groups are presented, taking into account the segment-level values for what concerns the features computed with openSMILE, as they capture more precise differences.

The "**Work**" narrative task (Table 4.11) contains 26 significant features, with the most discriminative being the low and higher order Mel-Frequency Cepstral Coefficients, speech and silence statistics, and the standard deviation at the bandwidth of the second formant (F2). We also observe features that were not significant when performing the analysis across all tasks, namely the standard deviation of the first formant bandwidth (F1) which is lower for the MCI group, and the segment-level self-similarity measure (HFD with the largest window), which demonstrates a lower mean and variability.

The amount of significant features considerably drops at the **picture description** task ("**Picture**"), which contains 15 relevant measures (Table 4.10). These are limited to solely two categories of the acoustic feature set; statistics from speech and silence duration, and the spectral features - lower order MFCCs (1-3) and the standard deviations of MFCC 7 and 8, as well as spectral skewness (standard deviation) and psychoacoustic sharpness (mean value).

The "**Dream**" Task is also limited compared to the working day narration, containing speech and silence features, mean values of the first 3 MFCCs, and both means and standard deviations of spectral skewness and kurtosis. Interestingly, we observe that it captures the lower mean F3 frequency for the MCI participants which was lost in the "Picture" Task, albeit with a moderate significance (Table 4.12).

Evidently, the most prominent acoustic features that repeat throughout all three tasks are

the measures of speech and silence duration, and Mel-Frequency Cepstral Coefficients, as well as spectral descriptors (particularly skewness and kurtosis).

Features lost between the tasks are certain silence measures (the average silence count at the picture description task), while the dream task lacks higher order MFCCs and instances of spectral centroid and psychoacoustic sharpness, showing however considerably lower values of kurtosis and skewness for the MCI group, which is not apparent at the "Work" task.

Additionally, regarding formant variables, differences are more prominent in the "Work" task, which captures lower oscillations at the frequency bandwidths of all of the first three formants, (F1, F2 and F3), followed by the "Dream" task, where we note a lower mean frequency at the third formant for the MCI group, similarly to the analysis which considered the combination of the linguistic tasks. In the picture task however, the relevance of any formant feature is lost.

The acoustic correlates of voice quality also decrease in relevance; the only relevant acoustic feature in this set is the higher variability of the Noise to Harmonics Ratio, captured at the work narration task. Lastly, the variables of RMS energy (prosodic), spectral variance and flux (spectral), and shimmer (voice quality), that demonstrated moderate relevance when cumulatively evaluating the tasks from all samples, are not captured in any of the single linguistic tasks.

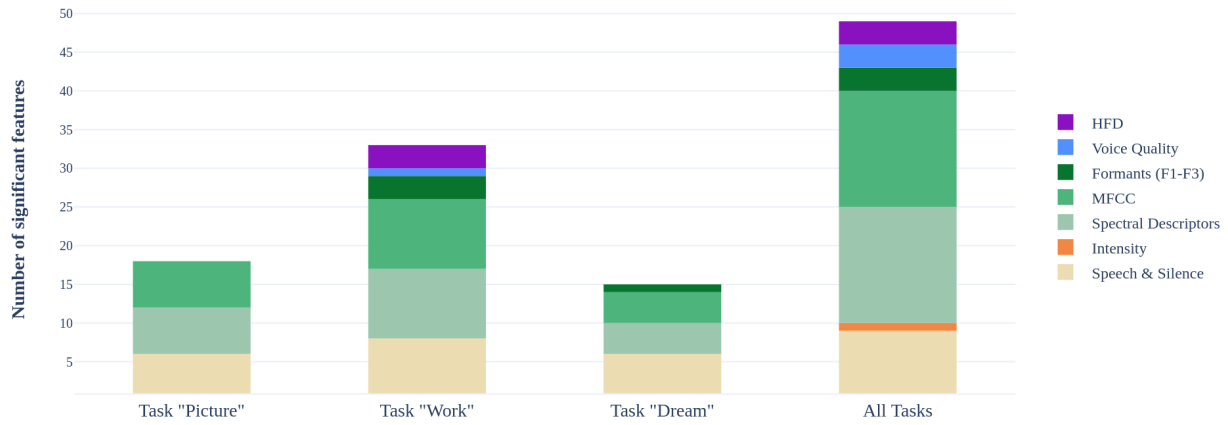


Figure 4.4: Significant features for each task.

Task "Picture"	CON_mean	CON_std	MCI_mean	MCI_std	P-Value	Signif.
SPE_ms_std	-0.217	0.024	-0.230	0.010	0.001	***
mfcc[2]_mean	0.296	1.099	-0.324	0.721	0.001	***
T_phon_rate_segment	0.234	0.949	-0.322	1.051	0.004	**
mfcc[3]_mean	0.327	0.985	-0.366	0.877	0.004	**
SIL_ms_std	-0.206	0.018	-0.204	0.021	0.008	**
mfcc[1]_std	0.240	0.910	-0.106	0.911	0.019	**
SPE_ms_mean	-0.205	0.047	-0.225	0.017	0.022	*
spectralSkewness_std	0.317	1.098	-0.279	0.648	0.022	*
SIL_to_SPE_mean	-0.320	0.516	0.440	1.349	0.022	*
psySharpness_mean	-0.280	0.936	0.294	0.989	0.022	*
T_phon_rate	0.391	0.748	-0.308	0.976	0.022	*
mfcc[8]_std	0.203	0.900	-0.111	1.027	0.025	*
SIL_ms_mean	-0.216	0.019	-0.214	0.022	0.025	*
mfcc[7]_std	0.290	0.932	-0.242	0.947	0.034	*
mfcc[1]_mean	0.259	0.938	-0.288	1.017	0.045	*

Table 4.10: Significant Features for the "Picture" Task at the segment level.

Task "Work"	CON_mean	CON_std	MCI_mean	MCI_std	P-Value	Signif.
mfcc[3]_std	0.267	0.975	-0.186	0.890	<0.001	***
F2bandwidth_std	0.203	0.958	-0.166	1.000	<0.001	***
SIL_to_SPE_mean	-0.320	0.516	0.440	1.349	<0.001	***
mfcc[10]_std	0.202	0.901	-0.116	1.047	<0.001	***
SPE_ms_mean	-0.205	0.047	-0.225	0.017	<0.001	***
mfcc[3]_mean	0.327	0.985	-0.366	0.877	<0.001	***
T_phon_rate	0.391	0.748	-0.308	0.976	0.001	***
SIL_count	-0.317	0.857	0.435	1.088	0.001	***
spectralCentroid_mean	-0.261	0.923	0.280	1.019	0.003	**
SPE_ms_median	-0.165	0.080	-0.191	0.025	0.003	**
psySharpness_mean	-0.280	0.936	0.294	0.989	0.004	**
logHNR_std	0.199	0.874	-0.035	0.942	0.004	**
SIL_ms_std	-0.206	0.018	-0.204	0.021	0.004	**
mfcc[2]_mean	0.296	1.099	-0.324	0.721	0.005	**
mfcc[7]_std	0.290	0.932	-0.242	0.947	0.008	**
SPE_ms_std	-0.217	0.024	-0.230	0.010	0.008	**
SIL_ms_mean	-0.216	0.019	-0.214	0.022	0.009	**
F1bandwidth_std	0.165	1.071	-0.056	0.802	0.015	*
mfcc[8]_std	0.203	0.900	-0.111	1.027	0.017	*
F3bandwidth_std	0.075	0.882	-0.048	1.153	0.022	*
mfcc[1]_mean	0.259	0.938	-0.288	1.017	0.022	*
HFD_w1280_mean	0.030	1.201	0.020	0.695	0.027	*
mfcc[1]_std	0.240	0.910	-0.106	0.911	0.029	*
mfcc[2]_std	0.237	1.040	-0.175	0.849	0.045	*
HFD_w1280_std	-0.004	1.185	-0.096	0.654	0.045	*

Table 4.11: Significant Features for the "Work" Task at the segment level.

Task "Dream"	CON_mean	CON_std	MCI_mean	MCI_std	P-Value	Signif.
SPE_ms_std	-0.217	0.024	-0.230	0.010	0.001	***
SPE_ms_mean	-0.205	0.047	-0.225	0.017	0.001	***
SIL_count	-0.317	0.857	0.435	1.088	0.002	**
spectralKurtosis_std	0.299	1.168	-0.299	0.568	0.003	**
mfcc[2]_mean	0.296	1.099	-0.324	0.721	0.003	**
spectralSkewness_std	0.317	1.098	-0.279	0.648	0.006	**
T_phon_rate_segment	0.234	0.949	-0.322	1.051	0.016	*
T_phon_rate	0.391	0.748	-0.308	0.976	0.017	*
mfcc[3]_mean	0.327	0.985	-0.366	0.877	0.020	*
mfcc[1]_mean	0.259	0.938	-0.288	1.017	0.023	*
spectralKurtosis_mean	0.277	1.171	-0.304	0.608	0.031	*
SIL_to_SPE_mean	-0.320	0.516	0.440	1.349	0.031	*
spectralSkewness_mean	0.258	1.135	-0.289	0.691	0.034	*
F3frequency_mean	0.179	1.088	-0.215	0.851	0.042	*
mfcc[5]_std	0.198	0.952	-0.139	1.014	0.048	*

Table 4.12: Significant Features for the "Dream" Task at the segment level.

4.2.3 Evaluation of acoustic features for disease progression

The last analysis includes the examination of how the relevance of the acoustic variables chosen might change when assessing their statistical significance using the speech samples from the entire corpus, including the subset of the participants of the pathological group that had an early dementia diagnosis (16 subjects, described at Section 3.2). This particular evaluation can provide some insights into the potential of the selected features for what concerns monitoring disease progression, but more importantly, it may allow us to assess whether certain variables can be less sensitive to speech alterations associated with early stage MCI, by comparing them with the findings outlined in the previous section (Section 4.2.1.1).

Naturally, there is a common trend in the results of this section and the previous ones; the most significant features are information regarding speech and silence duration, spectral characteristics, Mel-Frequency Cepstral Coefficients, and formant frequencies and bandwidths. However, the differences between the two groups in certain acoustic correlates are amplified enough to be deemed significant when performing the KS test at the segment level values.

Table 4.13 shows the distribution of the prosodic features, where speech and silence durations become even more prominent. Interestingly, some of the other variables rise in significance, albeit very moderately. In detail, the pathological group values show a decreased mean RMS energy and a lower standard deviation of perceptual loudness, as well as a lower variability in pitch when measured in semitones, which was not observed in the previous analysis. Like in the previous examination, the linear frequency measures are the least relevant.

For what concerns the features computed from the spectrum of the signal, the spectral descriptors and the MFCC values become more discriminative, which was mostly expected given their high relevance in the previous analysis. Again, there is a consistent pattern in the speech of the pathological group that contains a higher spectral centroid, and lower values of skewness, kurtosis and variance. Additionally, in this comparison we notice a lower variability in the spectral flux in the speech of the pathological group, although no differences are found in the standard deviation values of sharpness, spectral centroid or spectral variance (Table 4.17). Moreover, formant alterations become particularly

prominent; the pathological group exhibits lower values the mean frequency of the third formant and standard deviation of the second formant bandwidth with a p-value of under 0.001 (Table 4.15), showing an impressive increase in significance compared to the previous analysis between the control group and the MCI subset only (4.5).

The voice quality correlates outlined in (Table 4.16), demonstrate that the samples of the pathological group contain a significantly diminished Harmonics to Noise Ratio, with less deviations. Interestingly, we also note substantially lower shimmer variations but no meaningful differences in its mean value, but like before, there are no notable differences in any of the jitter values.

To conclude, the fractal dimension measure also performs better in this analysis (Table 4.18). On average, there is a significantly higher signal complexity for what concerns the pathological group, even at smaller windows. Additionally, differences are captured when the Higuchi Fractal Dimension algorithm is applied at the segment level as well.

Prosodic features	CON_mean	CON_std	PG_mean	PG_std	p-value	Signif.
SPE_ms_std	0.371	1.262	-0.340	0.478	<0.001	***
SIL_to_SPE_mean	-0.422	0.444	0.387	1.198	<0.001	***
T_phon_rate	0.465	0.779	-0.427	0.996	<0.001	***
SPE_ms_mean	0.332	1.302	-0.305	0.425	<0.001	***
SIL_count	-0.461	0.758	0.423	1.013	<0.001	***
SPE_ms_median	0.268	1.349	-0.246	0.368	<0.001	***
SIL_ms_std	-0.260	0.965	0.238	0.979	<0.001	***
SIL_ms_mean	-0.258	0.896	0.237	1.038	<0.001	***
T_phon_rate_segment	0.264	0.928	-0.242	1.009	<0.001	***
SIL_ms_median	-0.205	0.783	0.188	1.138	0.014	*
Loudness_std	0.151	0.937	-0.139	1.042	0.029	*
F0semitone_std	0.157	0.997	-0.144	0.988	0.034	*
RMSenergy_mean	0.185	0.998	-0.170	0.978	0.043	*
RMSenergy_std	0.181	0.974	-0.166	1.001	0.058	
F0semitone_mean	-0.154	0.998	0.141	0.988	0.077	
Loudness_mean	0.043	0.886	-0.040	1.099	0.097	
F0final_mean	-0.152	0.965	0.140	1.018	0.132	
F0final_std	0.006	0.930	-0.006	1.066	0.580	

Table 4.13: Significance of prosodic features between healthy group (CON) and pathological group (MCI & e-D) computed on a segment level.

MFCC 1-14	CON_mean	CON_std	PG_mean	PG_std	P-Value	Signif.
mfcc[2]_mean	0.356	1.120	-0.327	0.745	<0.001	***
mfcc[1]_mean	0.352	0.925	-0.323	0.963	<0.001	***
mfcc[3]_mean	0.312	1.009	-0.286	0.909	<0.001	***
mfcc[8]_std	0.271	0.890	-0.249	1.036	<0.001	***
mfcc[7]_std	0.339	0.949	-0.311	0.949	<0.001	***
mfcc[10]_std	0.269	0.910	-0.246	1.022	<0.001	***
mfcc[1]_std	0.303	0.906	-0.278	1.007	<0.001	***
mfcc[3]_std	0.341	0.981	-0.312	0.918	<0.001	***
mfcc[2]_std	0.279	1.080	-0.256	0.851	<0.001	***
mfcc[4]_std	0.269	0.929	-0.247	1.005	<0.001	***
mfcc[6]_std	0.140	0.818	-0.129	1.132	<0.001	***
mfcc[11]_std	0.147	0.877	-0.135	1.090	0.001	***
mfcc[12]_std	0.112	0.826	-0.102	1.133	0.002	**
mfcc[5]_std	0.245	0.958	-0.225	0.991	0.002	**
mfcc[13]_std	0.152	0.830	-0.139	1.122	0.002	**
mfcc[9]_std	0.130	0.922	-0.119	1.059	0.006	**
mfcc[14]_std	0.090	0.833	-0.083	1.131	0.007	**
mfcc[10]_mean	0.142	1.049	-0.130	0.941	0.015	*
mfcc[12]_mean	0.215	1.024	-0.197	0.942	0.030	*
mfcc[6]_mean	0.070	0.793	-0.064	1.160	0.037	*
mfcc[11]_mean	-0.128	1.070	0.117	0.924	0.049	*
mfcc[13]_mean	0.113	0.947	-0.104	1.042	0.100	
mfcc[4]_mean	0.084	1.050	-0.077	0.953	0.108	
mfcc[8]_mean	-0.090	0.989	0.083	1.010	0.183	
mfcc[14]_mean	-0.043	0.952	0.039	1.047	0.373	
mfcc[5]_mean	0.064	0.916	-0.059	1.074	0.447	
mfcc[9]_mean	-0.033	1.052	0.030	0.956	0.651	
mfcc[7]_mean	-0.077	1.066	0.071	0.938	0.836	

Table 4.14: Significance of Mel-Frequency Cepstral Coefficients between healthy group (CON) and pathological group (MCI & e-D) computed on a segment level.

Formants	CON_mean	CON_std	PG_mean	PG_std	P-Value	Signif.
F2bandwidth_std	0.240	0.982	-0.220	0.972	<0.001	***
F3frequency_mean	0.268	1.057	-0.246	0.883	<0.001	***
F1bandwidth_std	0.166	1.133	-0.152	0.840	0.025	*
F2bandwidth_mean	-0.076	0.867	0.070	1.109	0.026	*
F2frequency_mean	0.084	1.074	-0.077	0.928	0.055	
F3bandwidth_std	0.086	0.904	-0.079	1.081	0.067	
F2frequency_std	0.119	0.924	-0.109	1.060	0.111	
F1frequency_mean	-0.103	0.968	0.095	1.026	0.128	
F3bandwidth_mean	-0.110	0.969	0.101	1.024	0.159	
F3frequency_std	-0.018	0.892	0.017	1.096	0.529	
F1bandwidth_mean	0.018	1.057	-0.017	0.952	0.712	
F1frequency_std	0.021	1.019	-0.019	0.989	0.987	

Table 4.15: Significance of formant features between healthy group (CON) and pathological group (MCI & e-D) computed on a segment level.

Voice Quality	CON_mean	CON_std	PG_mean	PG_std	P-Value	Signif.
logHNR_mean	0.208	0.909	-0.191	1.047	<0.001	***
logHNR_std	0.244	0.838	-0.224	1.087	0.001	***
shimmerLocal_std	0.218	0.908	-0.200	1.044	0.002	**
jitterDDP_std	0.147	1.069	-0.135	0.919	0.125	
jitterLocal_std	0.151	1.052	-0.138	0.936	0.113	
shimmerLocal_mean	-0.064	0.965	0.058	1.034	0.289	
jitterLocal_mean	-0.016	0.970	0.014	1.033	0.333	
jitterDDP_mean	0.004	0.944	-0.003	1.055	0.440	

Table 4.16: Significance of voice quality features between healthy group (CON) and pathological group (MCI & e-D) computed on a segment level.

Spectral Descriptors	CON_mean	CON_std	PG_mean	PG_std	P-Value	Signif.
spectralSkewness_std	0.372	1.142	-0.341	0.700	<0.001	***
spectralKurtosis_mean	0.351	1.217	-0.322	0.595	<0.001	***
spectralKurtosis_std	0.356	1.222	-0.327	0.580	<0.001	***
spectralCentroid_mean	-0.274	0.913	0.251	1.017	<0.001	***
psySharpness_mean	-0.276	0.930	0.253	1.002	<0.001	***
spectralSkewness_mean	0.323	1.153	-0.296	0.727	<0.001	***
spectralVariance_mean	-0.239	0.773	0.219	1.132	0.010	**
spectralFlux_mean	0.151	0.973	-0.139	1.011	0.044	*
spectralFlux_std	0.167	1.023	-0.153	0.961	0.073	
spectralVariance_std	0.134	1.050	-0.123	0.942	0.102	
psySharpness_std	0.061	1.083	-0.056	0.921	0.432	
spectralCentroid_std	0.056	1.063	-0.051	0.943	0.642	

Table 4.17: Significance of spectral features between healthy group (CON) and pathological group (MCI & e-D) computed on a segment level.

File level	CON_mean	CON_std	PG_mean	PG_std	P-Value	Signif.
HFD_w1280	0.281	0.811	-0.258	1.090	0.001	***
HFD_w640	0.267	0.818	-0.245	1.092	0.003	**
HFD_w360	0.204	0.853	-0.187	1.091	0.013	*
Segment Level	CON_mean	CON_std	PG_mean	PG_std	P-Value	Signif.
HFD_w1280_std	0.025	1.223	-0.023	0.747	0.009	**
HFD_w360_std	0.020	1.207	-0.018	0.771	0.010	*
HFD_w640_std	0.023	1.214	-0.021	0.760	0.012	*
HFD_w640_mean	0.020	1.208	-0.019	0.770	0.019	*
HFD_w1280_mean	0.019	1.235	-0.017	0.729	0.025	*
HFD_w360_mean	0.011	1.180	-0.010	0.808	0.234	

Table 4.18: Significance of the self-similarity measure (HFD_windowsize) between healthy group (CON) and pathological group (MCI & e-D).

4.3 Discussion and Limitations

In this work, the majority of the selected acoustic features demonstrated strong potential in distinguishing individuals in the early stages of cognitive impairment from healthy controls. However, the set of features that lack conclusiveness for the MCI group may offer directions for future studies. At the present chapter, the implications of our findings are discussed, namely the relevance of the acoustic features chosen in this thesis, followed by details on the elicitation task evaluation.

4.3.1 Predictive power of selected features

This section presents the relevance of our findings across previous studies, along with the existing limitations in the interpretability and accurate estimation of acoustic features of speech. Given the comparative nature of this work and most of the research devoted to the study the predictive potential of digital linguistic biomarkers, it is also important to consider parallel research on the linguistical effects of non-pathological ageing, for a concise evaluation of the reliability of each acoustic feature as a discriminative biomarker of cognitive deterioration.

4.3.1.1 Speech and silence features

The results confirm that when compared to their healthy counterparts, the speech of individuals affected by neurocognitive disorder exhibits a disproportional speech rate with hesitations that are higher in occurrence and longer in duration, even at the early phase of the illness. This corroborates the findings of numerous studies which highlight that impaired speech rate is one of the first linguistic symptoms of neurocognitive disorder. This group of variables captures symptoms of impaired cognitive processing that extend beyond memory recall to various cognitive domains that can affect language, manifesting in deficits in articulation, syntax, lexical access, and discourse planning (Boschi et al., 2017).

While these findings are very meaningful, a more detailed study should consider in *what context* pauses most frequently occur, or whether certain aspects of discourse can be associated with a longer delay in response. Such analysis could offer concrete insights into the functioning of different memory domains and identify patterns in the specific linguistic levels affected in the early stages of neurodegeneration. Furthermore, a more detailed

quantification of hesitations in speech must account for filled hesitations besides silent pauses; this requires the use of transcriptions, which were not integrated in this work as its main scope was to compute linguistic biomarkers relying solely on acoustic features.

4.3.1.2 Intensity and Pitch

The evaluation of the acoustic correlates of intensity and pitch, traditionally used in this domain along with speech rate for measuring impaired intonation, did not yield significant differences between subjects with MCI and healthy controls. Among these, the only relevant acoustic feature is that of the mean RMS energy (Table 4.2), suggesting some evidence of hypophonia in MCI speech, with no significant differences in variability within the utterances of two groups, similarly to Meilán et al. (2020) and the authors of the previous works who designed and evaluated this project's dataset (Calzà et al., 2021). On the other hand, the third analysis on the samples between the control group and the pathological group (including the subjects that have progressed to early dementia) does capture decreased variations in loudness and pitch (Table 4.13), though this cannot allow for generalizations given the small size of the samples.

While a substantial amount of research focuses on pitch alterations as potential digital biomarkers both in early and clinical stages of dementia, to this day the findings are varying. There are previous works that did not report considerable alterations in AD speech (Meilán et al., 2014; 2020), even when using variety of analyses (including minimum and maximum F0 values, percentiles, and autocorrelation estimates), contrary to other studies that noted pitch variability as an important distinguishing factor (Yamada et al., 2021).

While such differences can indeed be attributed to different pitch extraction algorithms, feature extraction techniques and possible imbalance between the speech samples obtained, it is also the particular nature of this acoustic feature that can make its accurate assessment more challenging. Overall, F0 variability has been reported to increase as part of healthy ageing which is attributed to an expected decrease periodic vocal fold vibrations and/or incomplete vocal fold closure (Gorham-Rowan & Laures-Gore, 2006), but a detailed review by Tucker et al. (2021) highlights that these changes may not be linear given that different studies have produced contradictory results, reporting both increases and decreases in pitch values in the speech of healthy subjects recruited for their studies. Moreover, F0 values may vary according to the individual's physical or mental state, which may be influenced

by many random factors such as the time of day during the interview task. It may also be possible that subjects will subconsciously or purposefully alter their tone depending on the setting, employing a professional use of voice (Teixeira et al., 2011; 2013).

To this end, there is a shift towards the study of impairments in affective prosody, given that evidence that certain neuropsychiatric symptoms - namely apathy, depression, and anxiety - can be indicative of dementia progression (Hu et al., 2020; Jang JY. et al., 2020). In detail, König et al. (2021) concluded that the relevance of pitch measures cannot be generalized as each of these neuropsychiatric symptoms can be associated with different prosodic patterns which also vary according to sex, while Haider et al. (2020), who trained an affect recognition system using subsets of the DementiaBank Pitt corpus³ and Emo-DB⁴ did not produce very significant classification results using features obtained from semi-spontaneous tasks from AD individuals. A different study by Oh et al. (2023) utilized speech elicitation tasks specifically designed for inducing emotional responses from the participants. The scope of that study was to identify prosodic variables that can distinguish individuals with dementia from subjects diagnosed with depression, and they managed to effectively isolate different patterns of impaired prosody between the two groups, including pitch. Thus, it may be possible that spontaneous speech tasks may limit the interpretability of these prosodic features.

Considering all the above, it seems that the reliability of pitch-based measures as reliable linguistic biomarkers of cognitive impairment cannot be generalized. We may acquire a more thorough understanding in studies designed for distinguishing linguistic and affective prosody, with the careful implementation of differently structured elicitation tasks. Additionally, a wider set of metrics should be integrated (such as percentiles, variation coefficients), in a standardized way that will allow precise comparison across findings, ideally applied to more confined age and sex groups.

4.3.1.3 Spectral descriptors

The spectral features implemented in this project are certainly effective in identifying speech deviations that can manifest in the earlier stages of cognitive impairment. A higher

³<https://dementia.talkbank.org/access/English/Pitt.html>

⁴<http://emodb.bilderbar.info/start.html>

concentration of energy in the central frequency of the spectrum is consistently reported in other studies (Themistocleous et al., 2020), while the set of spectral descriptors (skewness, kurtosis, variance and flux) is in line with observations of previous studies focused on both MCI and pre-clinical dementia (Fraser et al., 2016; Meilán et al., 2020; Yamada et al. 2021; Ivanova et al., 2022), confirming their potential as distinguishing voice markers.

4.3.1.4 Mel-Frequency Cepstral Coefficients

Similarly, this study reaffirms the discriminative power of perceptual acoustic features such as Mel-Frequency Cepstral Coefficients, similarly to previous works that used a variety of elicitation tasks (Alhanai et al., 2017; Fraser et al., 2016; Hernández-Domínguez et al., 2018; Meghanani et al., 2021, Yamada et al, 2021). Given their widespread use in clinical speech processing applications, their statistical importance in this study may not be surprising; however, it is very encouraging that their robustness in reflecting early linguistic signs of cognitive decline is once again confirmed.

In literature, decreased temporal variations in MFCCs are interpreted as markers of speech monotony and dullness in clinical descriptions (Cummins et al., 2013; 2015). Additionally, they may also suggest lower resonance in the vocal tract, thus reflecting impairments in articulatory precision (Fraser et al., 2016a; Al Hameed et al., 2019). Since these values represent the distribution of energy in different frequency bands of a signal, they are harder to interpret individually in the context of detecting cognitive decline. Lower order coefficients (MFCC 1-3) can be particularly affected by reduced loudness in speech, but this characteristic, along with reduced intonation and impaired voice quality may be reflected in all of the coefficients, including the middle and higher range, as they are a compact representation of the spectral envelope.

4.3.1.5 Formants

It is very interesting that prominent alterations in the second and third formant were found in this study on MCI, even when computed from semi-spontaneous speech samples in Italian. Previous findings reporting significant discrepancies in the second and third formants were in different languages -namely Spanish (Martinez-Sanchez et al., 2018, on AD speech) and Swedish (Themistocleous et al., 2018; using MCI samples produced with sustained vowel phonation). Other studies which did not document the statistical significance of their selected features, implemented formant variables for developing high

accuracy screening systems in Japanese (Kato et al., 2013; 2015), and with a multilingual corpus in Lopez-de-Ipina et al., 2017 (with samples in English, French, Spanish, Catalan, Basque, Portuguese, Chinese and Arabic).

Generally, formant frequencies alterate in speech production with changes in vocal tract configuration relating to tongue, lip and pharyngeal movements. In most languages, including Italian, each vowel is described by the joint and unique contribution of its first and second formant (Leoni et al., 2004); for example, the second formant F2 is influenced by anterior-posterior movement of the tongue: when the tongue moves forward to form the vowel /i/, it generates an increase in F2 and a decrease in F1, while F2 is decreased as the tongue moves backward to generate the vowels /u/ and /a/. The third formant (F3) is considered an acoustic correlate of lip rounding during speech production, and lip rounding will result in lower frequencies at the F3 (Vizza et al., 2018).

While changes in formant trajectories are reported as a part of healthy ageing (Tucker et al., 2021), neurodegenerative illnesses can cause a deterioration in the range of articulatory movements, causing lower values at higher formant frequencies, and higher values at lower formant frequencies. This phenomenon is known as vowel formant centralization, which is often studied in Parkinson's disease (Bang et al., 2021), and typically analyzed in research by using the Vowel Space Area (VSA), which constructs a bi-dimensional space in the F1 and F2 plane, based on the Euclidean distances between F1 and F2 coordinates of different corner vowels (Vizza et al., 2018).

Given the above, it is difficult to establish a clear etiology of formant alterations in this work, besides attributing them to impairments in articulatory coordination typically caused by neurocognitive disorder. Since formant frequencies are formed based on different vowels and consonant context, a more thorough investigation in Italian would necessitate highly local estimations, by conducting vowel level comparisons between subjects, ideally counteracting age, sex and cultural differences. This would certainly be a challenge considering the data scarcity of this domain; nevertheless, it is very promising that the predictive potential of these features remains robust even when derived from semi-spontaneous speech samples, which can be obtained more easily compared to studies that require a stricter design.

4.3.1.6 Voice quality

The voice quality measures employed in this study demonstrate that voice deterioration - indicated by higher shimmer and diminished harmonics to noise ratio - may appear even at the first signs of cognitive impairment, and become more prominent when taking into account early dementia, albeit with a moderate significance compared to what is reported in progressed stages of the disease (Lopez-de-Ipiña et al., 2015; Warnita et al., 2018).

Again, it is relevant to mention that altered voice quality patterns can be associated with healthy ageing, as anatomical and physiological changes in the larynx and vocal tract can typically manifest as increased jitter and shimmer levels, and a lower HNR (Ivanova et al., 2022; Martinez et al., 2021; Tucker et al., 2021), which may explain the moderate relevance of these features presented in the outcomes of this work. Nonetheless, this project's findings agree with Themistocleous et al. (2018), that observed elevated shimmer levels with no particular variations in jitter in MCI speech, and the study of Meilán et al. (2018) that compared samples between subjects with MCI and progressed Alzheimer's disease, concluding that voice quality deteriorates especially at the later stages (Meilán et al., 2018), with AD individuals exhibiting a higher shimmer, jitter, and harmonics to noise ratio.

The elevated shimmer and HNR values in the speech of MCI individuals suggests that cognitive deterioration can be associated with more severe dysphonia compared to non-pathological ageing, given that neurocognitive disorder can impact fine motor control or cause mass lesions in the vocal folds, causing air leakage and amplitude irregularities due to incomplete glottal closure (Vizza et al., 2018). Taking into account the comprehensive analysis performed in this work, there is reason to suggest that while acoustic correlates of voice quality may not be as robust as the other features reviewed for distinguishing MCI speech, they are certainly useful for monitoring the progression of pathological symptoms.

4.3.1.7 Fractal dimension

To conclude, the self-similarity feature as computed from the waveform of each sample using the Higuchi Fractal Dimension algorithm seems particularly relevant, especially when applied on the entirety of the utterance - on a "file" level - with a larger window of 1280 points (described in Section 3.3.4.3). This applies particularly to the speech of MCI

subjects utilized in this study, and it seems that performing the analysis on the entire corpus can capture differences in signal complexity using smaller windows, even when applied on speech segments of the utterance.

However, regarding the identification of MCI speech in particular, unlike the rest of the frame-based features whose final values were more meaningful when computed at the segment level, a global utterance computation seems more appropriate for this feature for two reasons. Firstly, the segment-level implementation of the HFD was notably more computationally intensive, and had a diminished significance compared to the global computation, which can be attributed to the fact that it was performing iterations on limited data, thus not effectively capturing the fractal dimension of the signal. Additionally, although its segment level values were proved as one of the significant features when assessing the "Work" linguistic task only (4.11), confirming its relevance would need a larger sample, as it is not safe to make assumptions given solely its performance at a limited subset of the corpus.

We must also consider the fact that this feature captures signal complexity in a composite way, which makes it challenging to assess its particular effectiveness in targeting speech alterations caused by cognitive impairment specifically; on the other hand, the fractal dimension approach can be an advantage when dealing with imbalanced datasets (Lopez de Ipiña et al., 2013), with the additional benefit of being simple to implement on the entire waveform. In every case, considering that its discriminative power is also reported in Calzà et al., (2021), and in the work of Lopez-de-Ipina et al. (2015) who used a multilingual corpus, there is evidence to suggest that it can effectively contribute to the classification of MCI speech when added in the acoustic feature set.

4.3.2 Elicitation tasks

The secondary analysis on the linguistic tasks utilized for the corpus used in this study highlights that a better representation of daily-life speech necessitates their combination, as each task alone will suffer from limitations, as it is also reported by Calzà et al. (2021).

These tasks involve different cognitive skills for prompting the semi-spontaneous linguistic production in the subjects, targeting semantic, autobiographical and episodic memory respectively, although all three require essential skills in recalling working

memory (of what is being said), episodic memory (what has already been said) and prospective memory (planning what will be said).

In particular, picture-description tasks may not always be sensitive for capturing language deficits in speakers in the early stages of cognitive impairment, as subjects may apply compensatory strategies concealing language deficits (Ivanova et al., 2022). The dream narration task is certainly useful for capturing memory recall deficits, but it may introduce greater cognitive demands even in healthy speakers too, possibly skewing the results.

In this study, the most successful elicitation task in terms of acoustic feature significance was the "Work" task, which is to be expected as it can more effectively approximate the naturalness of spontaneous speech. However, as observed previously, it is not as efficient in capturing certain features when utilized alone. Additionally, more spontaneous narrative tasks can be less effective for certain linguistic deficits compared to constrained linguistic tasks such as the picture description, which are overall thought as more appropriate for capturing lexical impairments (Boschi et al., 2017). Given the above, a comprehensive assessment must ideally include a combination of tasks to effectively approximate language impairments.

4.3.3 Limitations in clinical applicability

While it is very promising that the results of this study are validated by previous findings, certain features, particularly those related to prosodic measures of pitch and intensity, present challenges in terms of interpretability and reproducibility. This can be attributed to the variability in feature extraction approaches and linguistic tasks across studies. The generalization of each feature's predictive power becomes more challenging, when considering that it is difficult to assess whether the amplification of speech deterioration symptoms - for example, voice quality - can be also attributed to normal ageing; this is particularly problematic as dementia symptoms and subsequent linguistic alterations gradually deteriorate as with the age of the speaker.

In terms of linguistic tasks, obtaining semi-spontaneous speech is generally recommended given their resemblance to natural language usage in daily life, and because such samples are relatively easier to obtain in a larger scale (De La Fuente-Garcia et al.,

2021). At the same time, strictly designed linguistic tasks offer the advantage of standardized procedures and better control over potential confounding factors, which can allow for more reliable and reproducible results. For what concerns the effective study of prosodic features of pitch and intensity, current research suggests that a more meaningful investigation may need to distinguish linguistic and affective prosody when designing the corpus. This may require the use of structured elicitation tasks aiming to evoke emotional responses from the participants, unlike the corpus of this project, which is comprised of semi-spontaneous speech samples. Additionally, for a more detailed examination of formants, sustained phonation tasks (Themistocleous et al., 2018) or reading aloud tasks are suggested as more appropriate alternatives (Meilán, et al., 2018).

All the issues outlined above stress that a more thorough analysis on acoustic features requires more large-scale data from individuals with Mild Cognitive Impairment. An important limitation of this study stems from the small size of the dataset, particularly since the main analysis focused on the distinction of the samples belonging to the MCI subset of the corpus, which was smaller compared to the control group. While this was a necessary choice in order to accurately assess the linguistic alterations associated with early cognitive decline specifically, a fair comparison should ideally be performed on groups with equal number of samples. Lastly, a more comprehensive evaluation of certain acoustic features - particularly those of pitch related measures and formant information - requires more appropriate elicitation tasks as those suggested above, preferably applied to more confined age and sex groups.

With regard to the feature extraction process, it's essential to acknowledge that the selected acoustic feature set is not exhaustive. Throughout the development of this project, additional features and speech analysis methods were considered, but were unfortunately not integrated into the project due to the necessity of adhering to the initial objectives of this thesis, namely the evaluation of new categories of features by adding spectral descriptors, formants, and voice quality correlates on Italian samples, achieved through the development of a feature extraction pipeline.

The measures that were not selected are predominantly spectral representations and estimates of the fundamental frequency envelope using alternative pitch contour methods. While these were available with the openSMILE toolkit, they were applied in studies that

leveraged all of the features of standardized sets available with the openSMILE library (Luz et al., 2020), without explicitly documenting the unique contribution of each variable. Thus, as outlined in the methods of this study (Section 3.1.1), a selection process was imposed since conducting a comprehensive evaluation of all potential features would not have been feasible. Fortunately, the feature extraction pipeline can be easily customized, thus its further development will concentrate on integrating more features.

4.3.4 Future Developments

A further development would leverage the openSMILE toolkit once again, as it offers noise-robust methods, enhancing reliability. Furthermore, it can allow for cross-study comparisons, since it is an open source package which additionally provides a comprehensive documentation as presented in Eyben (2016).

In particular, the future development of the feature extraction pipeline will explore the following measures:

- Long Term Average Spectra, using PLP (Perceptual Predictive Coding), which is an extension to the standard LP analysis by a psychoacoustic model of sound perception, and
- RASTA (Relative Spectral Transform) filter spectra, that accounts for both temporal characteristics of human hearing and speech production systems. These are documented in Eyben, 2016, pp. 55-60, p. 261.

These features can be used either to represent the average long-time spectrum measured at specific frequency bands, or provide its overall representation (Eyben, 2016, pp. 261). While the implemented spectral descriptors effectively capture both static and temporal characteristics, adding the following features can potentially enhance the performance of future classifiers.

- Additionally, Perceptual Linear Prediction can also be leveraged to extract the corresponding Cepstral Coefficients (PLP-CC), documented in Eyben, 2016, pp. 55-57).

- Harmonic ratios, computed from the relative amplitudes of F0 harmonics: H1–H2 and H1–A3, (Eyben., 2016, p. 270), also used in Themistocleous et al., 2020;
- Harmonics to noise ratio using the autocorrelation method which may be less sensitive to environmental noise conditions (Eyben, 2016, p. 78). Since this measure was significantly different between the samples of the two groups, incorporating this alternative technique would serve for a better assessment of its relevance.

Additionally, further development of the pipeline should consider statistical functionals beyond the means and standard deviations of low-level descriptors. In particular:

- variation coefficients which can minimize the influence of mean values; used by Yamada et al. (2021) to estimate pitch variability,
- pitch percentiles, which are more robust to outliers compared to using the minimum and maximum values (Eyben, 2016, p. 90-91),
- higher order statistical moments (skewness and kurtosis) on all features (Eyben, 2016, pp. 84-95 p. 263)
- regression coefficients -spectral slope and offset, (documented in Eyben, 2016, p. 92), as well as their temporal distributions, namely mean and standard deviation of the slope of rising/falling signal parts for pitch and loudness (Eyben, 2016, pp. 97-100, pp.270).

Lastly, an alternative estimation of the F0 envelope can be implemented with the *emobase* standardized feature set set, which uses a hybrid autocorrelation/cepstral method (Eyben, 2016 pp. 64-72), while the default pitch detector employed in the other feature sets uses the components from the power spectrum.

Judging by the significance of MFCCs in this study and previous research, it seems promising to explore prosodic features using acoustic perception. This can be utilized using the openSMILE toolkit, and its configuration is documented in the openSMILE

repository.⁵ Alternatively, the cepstral/autocorrelation method can also be implemented independently, by creating customized config files, although this set offers the advantage of including multiple statistical functionals. While this feature set is typically used for emotion recognition, aforementioned studies that have used all of the openSMILE sets reported its great overall performance (Luz et al., 2020).

⁵<https://github.com/audeering/opensmile/tree/master/config/emobase>

5. Conclusions and Future Work

This thesis explored the relevance of acoustic features as digital linguistic biomarkers for the identification of individuals with Mild Cognitive Impairment (MCI) from healthy controls in Italian. The feature extraction pipeline was designed in order to ensure reliable measurements by leveraging the standardized feature sets available with the open-source openSMILE toolkit. The transparent documentation of the workflow implemented in this thesis encourages meaningful comparisons for future work, and the comprehensive analysis of the selected feature set in this study offers important insights for future directions.

The approach implemented in this thesis has effectively addressed its objectives. The proposed acoustic feature set demonstrates a high potential in the identification of early signs of neurocognitive disorder, providing a comprehensive set of features that can be integrated within the development of automatic classifiers. Additionally, the limitations addressed in this work offer important insights that can be considered by future studies aimed at refining the extraction of digital acoustic biomarkers and designing speech elicitation tasks.

While this project cannot answer to the barriers imposed by data scarcity in this domain, it encourages a further exploration of acoustic biomarkers. The validation of the findings of this project from non-Italian studies presents considerable evidence that the acoustic-based approach has the advantage of minimizing potential language-specific limitations. Furthermore, considering that transcription-free methods may facilitate the protection of data privacy, this may be a solution towards increased data availability in the future. Such advancements would greatly enhance the reliability of automatic screening systems for the early detection of neurocognitive disorders, ultimately transforming the ambition of fast and non-invasive diagnostic measurements into reality.

6. Appendix

Acoustic Feature	OpenSMILE Correlates	Details
Pitch (F0)	F0final_sma_mean, F0final_sma_std, F0semitoneFrom27.5Hz_sma3nz_mean, F0semitoneFrom27.5Hz_sma3nz_std	3.3.4.2
Intensity	pcm_RMSenergy_sma_mean, pcm_RMSenergy_sma_std, Loudness_sma3_mean, Loudness_sma3_std	3.3.4.2
Spectral Centroid	pcm_fftMag_spectralCentroid_sma_mean, pcm_fftMag_spectralCentroid_sma_std	3.3.4.2
Psychoacoustic Sharpness	pcm_fftMag_psySharpness_sma_mean, pcm_fftMag_psySharpness_sma_std	3.3.4.2
Spectral Skewness	pcm_fftMag_spectralSkewness_sma_mean, pcm_fftMag_spectralSkewness_sma_std	3.3.4.2
Spectral Kurtosis	pcm_fftMag_spectralKurtosis_sma_mean, pcm_fftMag_spectralKurtosis_sma_std	3.3.4.2
Spectral Variance	pcm_fftMag_spectralVariance_sma_mean, pcm_fftMag_spectralVariance_sma_std	3.3.4.2
Spectral Flux	pcm_fftMag_spectralFlux_sma_mean, pcm_fftMag_spectralFlux_sma_std	3.3.4.2
Mel-Frequency Cepstral Coefficients 1-14	mfcc_sma[n]_mean, mfcc_sma[n]_std (for n = 1 to 14)	3.3.4.2
Formants (F1-F3)	F1frequency_sma3nz_mean, F1frequency_sma3nz_std, F1bandwidth_sma3nz_mean, F1bandwidth_sma3nz_std, F2frequency_sma3nz_mean, F2frequency_sma3nz_std, F2bandwidth_sma3nz_mean, F2bandwidth_sma3nz_std, F3frequency_sma3nz_mean, F3frequency_sma3nz_std, F3bandwidth_sma3nz_mean, F3bandwidth_sma3nz_std	3.3.4.2
Jitter and Delta-Jitter	jitterLocal_sma_mean, jitterLocal_sma_std, jitterDDP_sma_mean, jitterDDP_sma_std	3.3.4.2
Shimmer	shimmerLocal_sma_mean, shimmerLocal_sma_std	3.3.4.2
Harmonics to Noise Ratio	logHNR_sma_mean, logHNR_sma_std	3.3.4.2

Table 1: All features implemented with openSMILE. "_sma": smoothing average window typically of 2 or 3 frames, "nz": values computed from non-zero frames, i.e. frames whose voicing probability is high enough.

Feature (File level)	CON_mean	CON_std	MCI_mean	MCI_std	P-Value	Signif.
SPE_ms_std	0.261	1.184	-0.359	0.492	<0.001	***
mfcc[3]_mean	0.299	1.008	-0.411	0.840	<0.001	***
psySharpness_mean	-0.248	0.925	0.341	1.008	<0.001	***
mfcc[2]_mean	0.254	1.093	-0.349	0.735	<0.001	***
spectralCentroid_mean	-0.240	0.899	0.330	1.046	<0.001	***
SPE_ms_mean	0.221	1.217	-0.303	0.446	<0.001	***
T_phon_rate	0.322	0.817	-0.442	1.067	<0.001	***
SIL_to_SPE_mean	-0.312	0.503	0.429	1.315	<0.001	***
SIL_ms_std	-0.185	0.979	0.254	0.983	<0.001	***
SIL_count	-0.309	0.835	0.424	1.060	<0.001	***
mfcc[1]_mean	0.234	0.910	-0.322	1.038	<0.001	***
spectralSkewness_mean	0.220	1.122	-0.302	0.710	<0.001	***
SPE_ms_median	0.174	1.249	-0.239	0.385	<0.001	***
mfcc[8]_std	0.130	0.959	-0.179	1.038	0.001	***
mfcc[3]_std	0.156	1.025	-0.214	0.934	0.002	**
mfcc[12]_mean	0.211	0.977	-0.290	0.968	0.002	**
T_phon_rate	0.228	0.925	-0.313	1.024	0.002	**
HFD_w1280	0.236	0.858	-0.324	1.096	0.003	**
mfcc[7]_std	0.221	0.986	-0.304	0.948	0.003	**
mfcc[10]_std	0.123	0.948	-0.169	1.054	0.005	**
SIL_ms_mean	-0.197	0.870	0.271	1.108	0.007	**
F3frequency_mean	0.147	1.084	-0.202	0.841	0.009	**
HFD_w640	0.211	0.875	-0.290	1.095	0.014	*
F2bandwidth_std	0.160	0.992	-0.220	0.979	0.014	*
RMSenergy_mean	0.178	1.010	-0.244	0.943	0.024	*
mfcc[13]_mean	0.187	1.025	-0.257	0.914	0.025	*
spectralVariance_mean	-0.163	0.786	0.225	1.208	0.026	*
mfcc[6]_mean	0.117	0.860	-0.160	1.155	0.026	*
F3frequency_std	-0.136	0.895	0.187	1.111	0.033	*
mfcc[2]_std	0.174	1.094	-0.239	0.807	0.033	*
HFD_w360	0.140	0.932	-0.193	1.066	0.038	*

Table 2: All significant features ($p < 0.05$) and statistical values across the control (CON) and MCI group for all tasks computed at the global/file level.

Feature (Segment level)	CON_mean	CON_std	MCI_mean	MCI_std	P-Value	Signif.
mfcc[2]_mean	0.356	1.120	-0.277	0.735	<0.001	***
SPE_ms_std	0.371	1.262	-0.290	0.524	<0.001	***
mfcc[3]_mean	0.312	1.009	-0.397	0.898	<0.001	***
SPE_ms_mean	0.332	1.302	-0.229	0.477	<0.001	***
T_phon_rate	0.465	0.779	-0.263	1.017	<0.001	***
SIL_to_SPE_mean	-0.422	0.444	0.232	1.159	<0.001	***
spectralCentroid_mean	-0.274	0.913	0.261	1.008	<0.001	***
SIL_ms_std	-0.260	0.965	0.173	0.969	<0.001	***
SIL_count	-0.461	0.758	0.204	0.962	<0.001	***
psySharpness_mean	-0.276	0.930	0.294	0.982	<0.001	***
mfcc[1]_mean	0.352	0.925	-0.188	1.003	<0.001	***
spectralSkewness_std	0.372	1.142	-0.248	0.674	<0.001	***
spectralSkewness_mean	0.323	1.153	-0.233	0.702	<0.001	***
SPE_ms_median	0.268	1.349	-0.178	0.415	<0.001	***
mfcc[7]_std	0.339	0.949	-0.203	0.965	<0.001	***
spectralKurtosis_mean	0.351	1.217	-0.253	0.632	<0.001	***
spectralKurtosis_std	0.356	1.222	-0.270	0.594	0.001	***
mfcc[10]_std	0.269	0.910	-0.053	1.057	0.001	***
T_phon_rate_segment	0.264	0.928	-0.280	1.028	0.001	***
mfcc[3]_std	0.341	0.981	-0.115	0.896	0.003	**
mfcc[8]_std	0.271	0.890	-0.039	1.016	0.005	**
F2bandwidth_std	0.240	0.982	-0.138	1.025	0.005	**
mfcc[6]_mean	0.070	0.793	-0.211	1.110	0.006	**
F3frequency_mean	0.268	1.057	-0.114	0.827	0.006	**
SIL_ms_mean	-0.258	0.896	0.224	1.141	0.007	**
mfcc[2]_std	0.279	1.080	-0.148	0.881	0.009	**
RMSEnergy_mean	0.185	0.998	-0.232	0.935	0.013	*
mfcc[13]_mean	0.113	0.947	-0.263	0.817	0.017	*
logHNR_std	0.244	0.838	0.020	0.903	0.018	*
mfcc[12]_mean	0.215	1.024	-0.257	0.946	0.018	*
spectralFlux_mean	0.151	0.973	-0.230	0.936	0.018	*
spectralVariance_mean	-0.239	0.773	0.155	1.162	0.024	*
mfcc[1]_std	0.303	0.906	-0.041	0.907	0.027	*
shimmerLocal_mean	-0.064	0.965	0.187	1.013	0.033	*
mfcc[10]_mean	0.142	1.049	-0.123	0.942	0.037	*
logHNR_mean	0.208	0.909	-0.001	0.963	0.041	*
mfcc[5]_std	0.245	0.958	-0.094	1.021	0.042	*
F3bandwidth_std	0.086	0.904	-0.040	1.181	0.044	*
mfcc[4]_mean	0.084	1.050	-0.115	0.944	0.046	*

Table 3: All significant features ($p < 0.05$) and statistical values across the control (CON) and MCI group, for all tasks computed at the segment level.

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